

SUPPLEMENTARY MATERIAL

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Supplementary Text

Supplementary Results and Discussion

The genome of WH8109

The genome of *Synechococcus* WH8109 is 2.1 Mbp with a G+C content of 60.1%. It has 2661 predicted protein coding genes and 53 RNA genes including 2 ribosomal operons and 44 tRNA genes. The *Synechococcus* WH8109 genome has 1881 and 1947 genes in common with *Synechococcus* WH7803 and WH8102, respectively, and 429-628 genes that are unique relative to each of these *Synechococcus* strains (Fig. 1, with E-value cutoff of $1e^{-7}$). A comparison of this genome to that of *Synechococcus* CC9605, the other sequenced *Synechococcus* strain belonging to clade II, shows that they have 2081 genes in common and that the majority of these two genomes are syntenous. A large inversion of 846 kb is found in the WH8109 genome in comparison to *Synechococcus* CC9605 and begins at base 628,018 of the *Synechococcus* WH8109 genome (Suppl. Fig. S3). Furthermore, the *Synechococcus* WH8109 genome has 11 hypervariable islands that are at least 8 kb long or contain 10 genes or more and an additional 8 smaller islets (Suppl. Table S13). These islands are positioned throughout the genome and contain a variety of genes unique to this strain (Suppl. Fig. S3, Table S13).

Characterization of the P-TIM40 phage

The P-TIM40 phage has a broad host range, infecting the low-light adapted *Prochlorococcus* strain NATL2A (the host of isolation), high-light adapted *Prochlorococcus* strains MED4 and MIT9515, and *Synechococcus* strains WH7803, WH8102 and WH8109. It forms plaques on the *Synechococcus* strains and *Prochlorococcus* NATL2A, but not on *Prochlorococcus* MED4 or MIT9515. P-TIM40 has a relatively long latent period of 10 h and a 14 h lytic cycle when infecting *Prochlorococcus* NATL2A.

Comparison of Syn9 infective phage yield and phage DNA production

Considerably more Syn9 phage DNA was replicated and released from infected cells than was incorporated into infective phage particles (Suppl. Fig. S4). Less phage DNA was released from *Synechococcus* WH8102 in comparison to the other two hosts, and fewer infective phage were released by *Synechococcus* WH8109 relative to *Synechococcus* strains WH7803 and WH8102 (Suppl. Fig. S4). Therefore, despite similar or higher levels of phage DNA released to the extracellular medium during infection of *Synechococcus* WH8109, fewer infective phage were produced on this host, indicating that a lower ratio of infective phage is produced on *Synechococcus* WH8109 than on the other two hosts.

Comparison between microarrays and RNA-seq measurements

The expression values from both the microarray and RNA-seq platforms are correlated in a linear manner for most of the intensity range. However, saturation of microarray-derived expression values can be observed for highly expressed genes (Suppl. Fig. S2A). This is especially noticeable for highly expressed phage genes, which show a much smaller dynamic range based on microarray measurements than based on RNA-Seq. For instance, at 1 h after infection, expression values of phage genes tend to be between 10 and 13 for microarrays (corresponding to an 8 fold range on the linear scale), whereas they tend to be between 8 and 16 for RNA-Seq (corresponding to a 252 fold range on the linear scale). In contrast, RNA-Seq data shows vanishingly low expression of many host genes at later stages of infection (1 and 2 h) (Suppl. Fig. S2A). This is likely to be a direct consequence of the striking dominance of phage transcripts (98%) in the total pool of transcripts (Fig. 1D), leading to an under-sampling of hosts transcripts in the RNA-Seq procedure. The microarrays better represent these genes, as hybridization is not affected by the high number of phage transcripts. Quantitative RT-PCR analysis of a select number of host and phage

genes confirmed that the transcript profiles of phage genes are well represented by the RNA-seq data and that host genes are well represented by the microarray data (Suppl. Fig. S2B, C).

Syn9 phage gene transcription

Clustering of phage genes: In order to assess the quality and gene composition of the number of clusters obtained by hierarchical clustering, we used the Jaccard coefficient as a stability measure. The most stable solutions were obtained when the phage gene expression profiles were divided into 3 or 4 clusters, as can be seen by the high frequency of high Jaccard coefficients for these numbers of clusters (Suppl. Fig. S6). The only difference between 3 or 4 clusters is the separation of 4 genes into a separate cluster. This fourth gene cluster consists of g184, g91, g92 and g93, which, on average, are expressed early and remain high at later stages of infection, but which display somewhat different expression patterns on the different hosts (Suppl. Fig. S5). Therefore, due to the similarly high confidence obtained for both $k = 3$ and $k = 4$ clusters, we decided that this low number of genes does not warrant a special gene cluster. When $k = 3$, these genes cluster with the early genes (Fig. 3, subcluster E3 in Suppl. Table S5).

tRNA genes: Five of the six tRNA genes transcribed with middle genes reflect large codon usage biases, i.e., are directed at codons common in middle and late phage genes but rare in the host genome ($p < 0.01$, Wilcoxon Signed-Rank test). These findings support previous predictions that the phage tRNA genes would enhance efficient translation of phage proteins during infection of *Synechococcus* hosts that have significantly different G+C content, especially the protein products of the late structural genes and some middle genes (Weigele *et al.*, 2007; Limor-Waisberg *et al.*, 2011; Enav *et al.*, 2012)

Bacterial-like genes: A group of bacterial-like genes in Syn9 form part of a gene cassette that appears to have been horizontally transferred among cyanophages (Sullivan *et al.*, 2010). This cassette includes the *petE* and *PTOX* photosynthetic electron transfer genes, the *gnd* and *zwf* pentose phosphate pathway genes, and an SNase-like endonuclease (g125) that is likely involved in nucleotide metabolism. This entire gene cassette was transcribed from both middle and late promoters, as was the third pentose phosphate pathway gene, *talC* and the Calvin cycle inhibitor, *cp12* (Suppl. Table S5). In our expression analysis these genes clustered with late genes, suggesting that they were primarily transcribed from late promoters. In comparison, a previous study found that the pentose phosphate pathway genes were expressed at the same time as middle genes and may have been primarily transcribed from the middle promoters (Thompson *et al.*, 2011). Differences in host growth and infection conditions exist between the studies: Here the hosts were grown with a 14 hour light period at a constant level of $30 \mu\text{mol photon}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ and were infected in the morning hours; whereas in the Thompson *et al.* (2011) study, the hosts were grown in a sunbox with gradually increasing and decreasing light levels with a maximum of $320 \mu\text{mol photon}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ for the 14 hour light period, were infected 2 hours prior to darkness and were then transferred to $50 \mu\text{mol photon}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ constant light until the end of the infection cycle. This suggests that under different host growth and infection conditions, these genes were transcribed from different promoters which may provide important flexibility for the timing of their transcription.

Complex expression patterns of phage genes: Complex expression patterns can result from expression of genes from multiple promoters. Alternative promoters can also dynamically shape the operon-structure landscape at different time points (Fig. 5B and Suppl. Fig. S10B). For example, gp214 and gp215 are transcribed as a bi-cistronic transcript from a middle promoter at 0.5 h after infection, but later become part of a larger polycistron (gp213-14-15) due to activation of an upstream late promoter (Fig. 5B). These results point to a simple mechanism for phages to generate complex expression patterns and modular operon structures from combinatorial usage of a small repertoire of promoter types. Indeed many genes in T4 (Miller *et al.*, 2003) and T4-like phages (Desplats *et al.*, 2002) are transcribed from multiple promoters.

For a number of genes alternative promoters were found within their ORFs. In these cases a single modular gene can give rise to multiple protein variants (Fig. 5C and Suppl. Fig. S10C). For example, at 0.5 h after infection the predicted endonuclease gp125 was transcribed from an internal middle promoter, presumably resulting in a protein variant that is 26 amino acids shorter than the full-length variant that is expressed from a late promoter (Fig. 5C). This is an elegant way for a phage to increase the coding capacity of its genome, as known for the baseplate and terminase genes in T4 (Nivinskas *et al.*, 1992; Franklin & Mosig, 1996).

Antisense phage transcripts: Some of the temporally regulated asRNAs may regulate the switch from middle to late transcription of the genes they overlap. For example, the asRNA overlapping the late genes gp130-131 is transcribed from a middle promoter at 0.5 h after infection, with no expression of the sense transcript. At 2 h after infection the asRNA disappears and expression of the sense transcript is driven by a late promoter (Suppl. Fig. S9A). This may represent an efficient mechanism to ensure silencing of a late gene at earlier time points where leaky expression (possibly via read through from an upstream promoter) might be harmful for the infection process.

Host gene transcription

tRNA genes: In addition to protein coding genes, levels of many tRNA genes remained unchanged and two, for arginine and alanine, increased significantly (Suppl. Table S14). No significant relationship was found between phage codon usage and the retention of high levels of these host tRNAs. Nonetheless, it is likely that indiscriminate maintenance of many host tRNAs would help ensure continued phage protein production throughout the infection cycle.

Mechanism of regulation in the broad group of T4-like bacteriophages

The broad group of T4-like bacteriophages can be divided into 3 diverse clusters based on phylogenetic analysis of structural proteins (Desplats & Krisch, 2003). Of these, the T4-like cyanophages are the most distant group relative to T4 and are categorized as the exo-T-even phages. We suggest that the mode of transcriptional regulation found in T4 may be limited to the closely related T-even phages, such as RB69 and 44RR (Nolan *et al.*, 2006). Indeed, phages in the wider pseudo-T-even group, such as RB49 and Aeh1, which infect *E. coli* and *Aeromonas hydrophila*, respectively, have bacterial σ^{70} -like promoters upstream of key DNA replication genes ((Desplats *et al.*, 2002); this study) similar to the cyanophages. However, they possess T4-like early promoters with UP elements (Nolan *et al.*, 2006) upstream of possible phage early genes, rather than the early motif we identified for the cyanophages. Conversely, we failed to find obvious σ^{70} like motifs upstream of either the replication or potential early genes in the marine *Vibrio* phage KVP40 (belonging to the schizo-T-even cluster), suggesting that it has a mode of regulation different to both T4 and the T4-like cyanophages. Thus the mechanism of regulation known so well for T4 may actually be the exception rather than the rule among the broader group of T4-like phages.

Supplementary Methods

Sequencing and analysis of the genome of *Synechococcus* WH8109

Synechococcus sp. strain WH8109 was grown from a single colony to approximately 10^{10} cells for genomic DNA extraction. The cells (concentrated by centrifugation) were embedded in a plug of low melting point agarose (SeaPlaque GTG, Lonza) and high molecular weight genomic DNA was extracted following previously described methods (Stein *et al.*, 1996). The genome was sequenced by the J. Craig Venter Institute by Sanger sequencing and assembled with the Celera v5.1 program. Five contigs resulted from the assembly. PCR and sequencing of the resultant fragments was used to close the gaps and

determine the order of the contigs, as well as to verify the location and orientation of a large inversion. The revised genome has been submitted to NCBI under the accession no. CP006882.

Prediction of ORFs was done using the GenemarkS program (<http://exon.gatech.edu/genemarks.cgi>) and additional ORFs predicted by the Glimmer program (<http://ccb.jhu.edu/software/glimmer/index.shtml>) were added. Annotation of predicted ORFs was carried out by homology comparisons using the conserved domain database (CDD) and the non-redundant database (nr) of the National Center for Biotechnology Information (NCBI) with BLASTP E-value thresholds of $<1e-5$. Gene names and product descriptions were taken from homologs in the WH8102 and CC9311 genomes in order to be consistent with gene identifiers from other marine cyanobacteria. The tRNAscan-SE program (Schattner *et al.*, 2005) was used to predict tRNA genes. Genes were numbered sequentially beginning with the *dnaN* gene. The revised genome has been submitted to NCBI under the accession no. CP006882.

Genomic islands or islets were defined as regions lacking synteny between WH8109 and other marine *Synechococcus* genomes, with islands consisting of regions of at least 8 kb or 10 genes (following Avrani *et al.* (2011)), whereas islets are smaller regions that encode at least one gene. These regions were identified in two ways. Firstly, regions that lack synteny with the genome of *Synechococcus* CC9605, the other sequenced clade II strain (Dufresne *et al.*, 2008). This was determined by regions of low similarity (E-value $>10e-5$) of at least a 2 kb region that were found using a 9 bp search window in the Geneious program using the EMBOSS tool dotmatcher. Secondly, islands that were defined by Dufresne *et al.* (2008) for CC9605 in comparison to 10 other *Synechococcus* genomes and are present in WH8109, were also defined as islands.

Isolation, characterization and genome sequencing of the P-TIM40 phage

The cyanophage P-TIM40 was isolated from the surface waters collected from the central Pacific Ocean off the Line Islands on 24 August 2007. The low-light adapted *Prochlorococcus* strain NATL2A was used as the isolation host. P-TIM40 infection dynamics were determined by one step growth curves using the plaque assay and qPCR of extracellular phage during infection of *Prochlorococcus* NATL2A, as described for Syn9.

Prior to genome sequencing of the P-TIM40 genome, the phage was grown using *Synechococcus* WH8102 as host. Phage particles were concentrated and purified on a CsCl gradient as described previously (Sabehi *et al.*, 2012). Phage genomic DNA was extracted using the Qiagen Lambda kit and sequenced using the Illumina Next-generation Genome Analyzer IIx.

Open reading frames were predicted using the GeneMark (<http://opal.biology.gatech.edu/GeneMark/>), MetaGeneAnnotator (<http://metagene.cb.k.u-tokyo.ac.jp/>), and Glimmer (<http://www.cbcu.edu/software/glimmer>) programs. BLASTP in the non-redundant database at the National Center for Biotechnology Information (NCBI) was used to search for sequence homologs of the translated ORFs with an upper threshold E-value of $1e-5$. Conserved domains searches were performed using NCBI Blast CDD search and Pfam (<http://pfam.sanger.ac.uk/>) to assist functional predictions. tRNAscan-SE program was used to identify tRNA genes. This genome has been submitted to NCBI under the accession number KP211958 and gene annotations can be found in Suppl. Table S8.

Quantitative PCR Protocol

Each qPCR reaction contained 1X Roche universal probe library (UPL) master mix (LightCycler® 480 Probes Master, Roche), 100 nM UPL hydrolysis probe, 200 nM of HPLC-purified primers and template in a total volume of 25 μ l (Primers and probes used appear in Suppl. Table S1). Reactions were carried out on a LightCycler 480 Real-Time PCR System (Roche). The cycling program consisted of an initial

denaturation step of 95 °C for 15 min followed by 45 cycles of amplification, each including 10 sec denaturation at 95 °C, and a 30 sec combined annealing and elongation step at 60 °C, at the end of which plate fluorescence was read (FAM; Ex/Em 465/510nm). The point at which the fluorescence rose above the background fluorescence (the crossing point or Cp) was used to determine the amount of initial target, using the absolute quantification/2nd-derivative maximum analysis from the LightCycler 480 software package (release 1.5.0). Quantification of template was determined from DNA standard curves of linear plasmid DNA containing the phage *g20* gene and genomic DNA for the host *rbcL* gene. Six different DNA concentrations were used for the standard curves and were chosen to span the levels found in the samples. These ranged from 10^6 to 10^{11} for the *g20* gene and from 10^3 to 10^8 for the *rbcL* gene. The R^2 value for all standard curves was greater than 0.99.

RNA extraction and DNase treatment

Total RNA was extracted from infected and control cultures using the PGTX method of Pinto *et al.* (2009) with minor modifications. Briefly, samples were frozen in the PGTX buffer containing 40 ml liquid phenol 95%, 6.9 ml glycerol, 0.1 g 8-hydroxyquinoline, 0.58 g EDTA, 0.8 g sodium acetate, 0.5 g guanidine thiocyanate and 4.6 g guanidine hydrochloride at final pH of 4.2 in a final volume of 100ml but without Triton X-100. Samples were thawed on ice and subjected to a 5-min incubation at 95 °C followed by a further 5-min on ice. The aqueous phase was extracted with an equal volume of chloroform:isoamyl alcohol (24:1, vol/vol). RNA was precipitated with 1 volume of isopropanol and washed in 75% ethanol. Total nucleic acids were quantified based on absorption at 260 nm and RNA integrity was verified by gel electrophoresis.

DNA was removed by DNase I digestion using the Turbo DNA-free kit (Ambion). For microarray analysis 15 µg of the nucleic acid extract was digested with 6 U of Turbo DNase during a 60 min incubation at 37 °C followed by DNase I inactivation with inactivation slurry. The RNA was purified and concentrated by sodium acetate/ethanol precipitation. DNA removal was verified by gel electrophoresis. For reverse transcriptase (RT)-PCR analyses DNA was removed from 1 µg of the nucleic acid extract with 6 U of Turbo DNase using the above procedure but without the precipitation step. DNA removal was verified by no-RT controls followed by qPCR for each sample.

Microarray experimentation

Design of microarray: An Affymetrix high-density custom-made tiling array was designed with 5 micron features. This array contains probes for the genomes of 3 *Synechococcus* strains (WH7803, WH8102 and WH8109) as well as 8 cyanophages (Syn9, Syn5, S-TIP37, S-TIM5, Syn19, S-TIM4, P-TIM40 and P-GSP1) and was named the SYN-PHG (SYN-PHGa520786F) array. The arrays contain probes that are 25 nucleotides long and only perfect match (PM) probes were included on the array. The probes were tiled across both strands for all of the genomes. The density of probes for the 3 *Synechococcus* strains was a probe centered every 35 nt with a 5 nt offset allowed, such that there was an average gap of 10 nt between probes. The probe density for the cyanophage genomes was every 17 nt with a 5 nt allowed offset, so that the probes overlap across the length of the genomes. Additional probes were designed for short ORFs such that a minimum of 8 probes were defined per ORF. Probes with similarity to rRNA genes were excluded (hard pruned) to prevent cross-hybridization with transcripts from these highly expressed genes. The array contains the following sets of Affymetrix control probes: *B. subtilis* probes for the *dap*, *lys phe* and *thr* genes that serve as labeling controls (known as poly A controls); the *E. coli* *bioBCD* and P1 phage *creX* probes that serve as hybridization controls and antigenomic probes with differing G+C content that serve as background controls. Two types of library files were generated for this array. The first is a CDF library file defining the probe sets for calculating expression summaries for all annotated open reading frames. The second is a BMAP (binary probe mapping) library file that contains information of the individual probes included on the array.

Sample preparation: Synthesis of cDNA, labeling, hybridization, staining and scanning was carried out according to Affymetrix protocols for Prokaryotic target RNA (GeneChip Expression Analysis Technical Manual, Prokaryotic; 702232, Rev.3) with minor modifications. Two micrograms of total RNA, together with the *B. subtilis* labeling controls (*lys* 19 pM, *phe* 38 pM, *thr* 76 pM, and *dap* 285 pM), were denatured at 70 °C for 10 min and annealed to random hexamer primers (25 ng/μl) at 25 °C for 10 min. The RNA was reverse transcribed to produce cDNA with 25 U/μl Superscript II (Invitrogen Life Technologies) and 0.5 mM dNTPs in the presence of 0.5 U/μl SUPERase•In RNase Inhibitor (Invitrogen) and 10 mM DTT. The mix was incubated at 25 °C for 10 min followed by 60 min incubations at 37 °C and 42 °C respectively. Superscript II was inactivated with a 10 min incubation at 70 °C. Sodium hydroxide (0.25 N) was used to remove RNA during a 30 min incubation at 65 °C, followed by neutralization with HCl. The cDNA was purified with MinElute PCR purification columns (Qiagen). Fragments of cDNA, 50-200 nt long, were produced from a 10 min incubation at 37 °C with DNase I (0.6 U per μg cDNA), followed by heat inactivation of the DNase I enzyme (10 min at 98 °C). The cDNA fragments were end labeled using GeneChip® DNA Labeling Reagent during a 60 min incubation at 37 °C. The reaction was stopped by the addition of 30 mM EDTA.

The cDNA was hybridized to the SYN-PHG custom Affymetrix array in aqueous hybridization solution (1X Affymetrix Hybridization mix, 7.8 % DMSO, containing the hybridization control mix: *bioB* (1 pM), *bioC* (0.05 nM), *bioD* (0.25 nM) and *cre* (1 nM) and 0.05 nM of the prelabeled Affymetrix hybridization B2 oligo control) during a 16 h incubation at 45 °C in a GeneChip Hybridization Oven 640 rotating at 60 rpm. Washes and stains were carried out with the GeneChip Hybridization, Wash and Stain Kit (Affymetrix) on a GeneChip Fluidics Station 450 (Affymetrix) following the FS450_0002 Affymetrix protocol. The arrays were scanned with the GeneChip Scanner 3000 7G (Affymetrix) at a 3.5 μm resolution with excitation set for 570 nm.

Microarray data analyses

For Syn9 infection of the three *Synechococcus* hosts, the dataset consisted of 121 microarray measurements with three biological replicates for each time point after phage addition (2-5 min and 0.5, 1, 1.5, 2, 3, 4 h) of infected and uninfected control cultures. A few samples were lost during preparation, resulting in 42, 39 and 40 microarrays for WH7803, WH8102, and WH8109, respectively. P-TIM40 gene expression was determined from microarray analyses. For P-TIM40 infection of *Prochlorococcus* NATL2A, the dataset consisted of 12 microarray measurement of 4 time points (0.5, 1, 2, 3 h) after infection, carried out in three biological replicates.

Probe signal intensities were read from CEL files and evaluated on the basis of the CDF library file using R statistical language with various Bioconductor add-on packages; *affy*, *AffyTiling*, and *limma*. Following previous analyses (Lindell *et al.*, 2007) background signal correction, normalization and summarization were performed by Robust Multiarray Averaging (RMA) from the *affy* package. Differential expression between infected and uninfected controls was statistically evaluated by a linear model comparing the time-resolved signal intensities from the arrays with uninfected hosts to those from arrays with infected hosts (*Limma* package; Smyth, 2005). *P*-values were converted to false discovery rates (*q*-values) by the Benjamini-Hochberg approach.

Determination of host gene expression groups (Suppl. Tables S9-S12): Host genes were sorted into four expression groups based on the microarray results as follows: (i) Genes with an immediate but transient increase in transcript levels after infection are those with a Log₂FC (infected/uninfected) > 0.5 and an adjusted *p*-value of *q*<0.05 at the first sampling point of 2-5 minutes after phage infection, that declined thereafter (Suppl. Table S9); (ii) Immediately declined host genes are those with a Log₂FC < -0.5 and *q*<0.05 at the first sampling point (Suppl. Table S10); (iii) Host genes with a subsequent increase in

transcript levels are those with a $\text{Log}_2\text{FC} > 0.5$ and $q < 0.05$ from the 30 min time points onwards (Suppl. Table S11); (iv) Host genes with unchanged transcript levels during the first 2-4 h after infection are those with a Log_2FC that did not change significantly ($q > 0.05$) for at least the first 2 h after infection (Suppl. Table S12). Unchanged host genes were only included if their transcript levels were above background levels and therefore, can be considered as expressed. Background levels were determined by comparing the signal intensity for these genes from one strain in uninfected control arrays for that strain (strain 1) to the signal intensities of these same genes from strain 1 in control experiments in which the other two host strains were investigated. Only genes for which log_2FC (strain1/other 2 strains) > 0.5 and $q < 0.001$ were considered to be expressed.

Conserved domain searches for differentially expressed host genes that were previously annotated as "hypothetical" proteins were performed using NCBI Blast CDD search and Pfam (<http://pfam.sanger.ac.uk/>). Determination of homologous genes among the three *Synechococcus* hosts as well as for *Synechococcus* genes homologous to *Prochlorococcus* MED4 upregulated host genes in response to P-SSP7 infection (Lindell *et al.*, 2007) was done using a reciprocal best BLAST hit approach using BLASTP at the National Center for Biotechnology Information (NCBI). In two cases of response genes that belong to gene families, we also identified homologous genes that do not necessarily return reciprocal best blast hits. This was for the porin genes and for the DUF1651-containing genes.

RNA-seq experimentation

Sample preparation: Each RNA sample was treated with TURBO DNase (Ambion) to digest residual DNA. RNA samples were analyzed using the 2100 Bioanalyzer (Agilent) to assess quality and treated with MICROBExpress™ Bacterial mRNA Purification Kit (Ambion) to remove rRNA from total RNA (He *et al.*, 2010). The efficiency of the rRNA removal was determined using the Bioanalyzer (Agilent) and if residual rRNA was observed, treated again with the MICROBExpress™ Kit. Samples were stored at -80°C .

Library preparation for 5'-end sequencing: Tobacco Acid Pyrophosphatase (TAP) treated libraries (see below) were prepared from two replicates of the same samples used in the whole-transcriptome libraries for time points 0.5, 1 and 2 h and the uninfected controls. Untreated libraries were similarly prepared for one replicate of each infected host, and for pooled triplicates of the uninfected hosts WH7803 and WH8109 (control) at 0.5 h time point. For each sample, 1.5 μg of rRNA-depleted RNA was incubated with or without 5 U of TAP for 2 h at 37°C to generate 5' monophosphate RNAs (TAP+ and TAP-samples, respectively). The 3' ends were blocked with NaIO_4 (10 mM) for 1.5 h at 4°C in the dark and saturated by adding 1/10 volume of 1 M lysine for 10 min at room temperature. Each reaction was passed through NucAway spin columns (Ambion). Illumina 5' adapters (in single-strand RNA form), with addition of a unique 4-base index sequence, were ligated to the resultant RNA using 25 units of T4 RNA ligase (Epicentre) in the presence of 40 units of RNase inhibitor (Promega) in a final volume of 50 μL . The reaction was incubated at 37°C for 2 h and precipitated with 1/10 volume of 5 M NH_4OAc and 100% ethanol, at 13,000 rpm for 15 min at 4°C . The ethanol was removed after chilling at -80°C for 1 h. The RNA pellet was washed twice with 70% ethanol, resuspended in RNase-free water and washed twice with ultra-pure water on YM-30 microcon columns (Millipore). First strand cDNA was synthesized using random hexamer primers attached to the Illumina 3' adapter with the SuperScript First Strand Synthesis system (Invitrogen). Adaptors and primer dimers were removed from the resultant cDNA using a PCR purification kit (Qiagen). The cDNA was then PCR amplified using primers matching the 5' and 3' Illumina adapters for 20 cycles. Fragments 250-500 bp long were extracted from 3% low melting point agarose gels and quantified with the Bioanalyzer. cDNA libraries of 2-10 subsamples were mixed in equal amounts and sent for Illumina sequencing.

RNA sequencing analyses

Read mapping: Reads of 41 to 51 nt were aligned separately for each sample to the reference genomes of the Syn9 cyanophage (Genbank: NC_008296) and the matching host (Genbank: NC_009481, NC_005070 and CP006882 for WH7803, WH8102 and WH8109, respectively). The reads were mapped using Novoalign 2.08.01 (Novocraft Technologies Sdn Bhd, <http://www.novocraft.com>) with the default parameters and [-c 1 -r Random]. For the mapping of the bisulfite sequencing reads, 100-nt long paired-end reads were aligned to the relevant reference genomes. The reads were mapped using Novoalign 2.08.01 with the parameters [-b] for the indexing step, and [-r Random] for the alignment. Reads that mapped to more than one position were discarded from further analyses.

Expression calling of phage genes: For each whole-transcriptome sequencing sample, reads covering each gene were counted, and normalized by gene length and by the number of reads mapped to the RNA component of the RNase P gene (*rnpB*) from the respective hosts.

Determination of phage TSSs: Phage TSSs were determined as previously described (Wurtzel *et al.*, 2012a) with some minor modifications. Briefly, TSS determination relies on the enrichment of TSSs within libraries treated by the Tobacco Acid Pyrophosphatase (TAP), which specifically cleaves primary 5' triphosphate containing ends, compared to untreated ones, which provides both primary and processed 5' ends. To differentiate between reads initiating from TSSs versus RNA processing sites, a set of 5' ends most likely representing real TSSs (positive set) was defined and compared with a group of 5' ends most likely representing processed sites (negative set). For each time point, the positive set included the 5' ends located upstream of genes (up to 100 nt) that were reproducible in the phage while infecting all 3 hosts. If multiple genomic positions were found for a gene within this distance, the position with the highest number of supporting reads was selected as the TSS for the positive set. Additional 5' ends farther than 100nt upstream of genes, for which a sharp increase (>5 fold) in the measured expression produced from the whole-transcript sequencing was observed, were also added to the positive set. The negative set was defined as genomic positions located more than 150 nt inside ORFs.

The TAP enrichment threshold for a TSS was calculated for each sample separately based on the distribution of the TAP+/TAP- ratio in the positive and negative sets (Wurtzel *et al.*, 2012a). A genomic position was considered a real TSS at a specific time point, if it was above the TAP+/TAP- threshold in all three hosts or when found in two hosts for multiple time points. Also, positions supported by a sharp increase in coverage were added.

The dominant TSS upstream of each gene (up to 400 bp) at each time point was considered to be that gene's primary TSS. TSSs internal to ORFs were associated with the gene only if they were supported by an increase in coverage, affecting the expression level of that gene. When no TSS was associated with a gene at a certain time point the gene was considered to be part of an operon with the upstream gene under the condition that it was both positioned on the same strand and expressed at the same levels (less than 2 fold change in the median coverage).

Computational identification of promoter motifs: The sequences upstream of TSSs (100 nt) associated with phage genes were extracted from the Syn9 genome. The sequences upstream of operons already transcribed at 5 min clearly showed the early motif by manual inspection. The remaining promoters sequences were grouped by time point of their first appearance (0.5 h, 1 h or 2 h). Motif discovery was run for each group using oligo-analysis and pattern-assembly from the Regulatory Sequence Analysis Tools (RSAT) web interface (<http://rsat.ulb.ac.be/>) with default parameters and [-1str]. Motif refinement was done using matrix-scan (Turatsinze *et al.*, 2008) with the parameters [-pseudo 1 -decimals 1 -1str -bginput -markov 0 -bg_pseudo 0.01 -lth score 1] as previously described (Wurtzel *et al.*, 2012b). Only matches positioned ± 1 nt or ± 2 from the main location of the -10 and -35 motifs, respectively, were

retained (Harley & Reynolds, 1987; Lisser & Margalit, 1993). If overlapping matches were found for a single promoter, the hit with the highest score was retained. For each motif, a new position-specific scoring matrix (PSSM) was built based on the hits and the entire process was performed iteratively until the matches converged (3 rounds). For the logos presented in Fig. 4C-D, the promoter sequences were aligned using ClustalW2 (Larkin *et al.*, 2007) separately for each motif (early motif box, -10 box, -35 box, TSSs) with manual adjustments. Logos were prepared using weblogo (Crooks *et al.*, 2004). The motifs were then searched for in sequences upstream of TSSs found in the middle of ORFs or in the antisense orientation (see *Syn9 transcriptome browser*, http://www.weizmann.ac.il/molgen/Sorek/syn9_browser/). Cases of discrepancy between cluster assignment and promoter class were examined manually. For searching the early motif in additional cyanophages, the sequences upstream of all ORFs of 15 cyanophages were extracted (150 nt) in addition to all intergenic sequences. The early motif was searched using matrix-scan (Turatsinze *et al.*, 2008) with a PSSM that was built based on Syn9 cases and the parameters [-pseudo 1 -decimals 1 -1str -bginput -markov 0 -bg_pseudo 0.01 -lth score 10 -uth pval 1e-5]. A similar search was done on the GOS (<http://camera.calit2.net>) dataset downloaded from NCBI (Accession number AACY020000000, sequences AACY020000001 - AACY024124495). To calculate the enrichment of phage genes among the contigs in which the early motif was identified, those GOS contigs were searched for phage genes using blastx against the Phage Orthologous Groups (POGs) dataset (Kristensen *et al.*, 2011, 2013) with E-value < 1e-20. The same search was performed on 10 sets of contigs, of the same total size, that were randomly selected from the GOS data. For Fig. 6B, ORFs predicted by Glimmer (Salzberg *et al.*, 1998) were added for each contig, on top of the viral hits from the POG dataset. Viral hits were also found to be homologous to either Syn9 early genes or putative early genes from other sequenced cyanophages.

asRNA analysis: TSSs that were in the opposite orientation of annotated ORFs were considered to be potential 5' ends of asRNAs. Potential ORFs downstream of these positions were tested using BLASTP for homology with known proteins (E-value <= 1e-4) but no such homologies were found. The transcript length of asRNAs was estimated by scanning the read coverage produced by the strand-specific bisulfite sequencing, downstream of the reproducible site. A sharp decline in the read coverage was considered the end of the transcript (Suppl. Fig. S15).

Comparing phage transcription patterns between the different hosts

To examine whether the temporal expression of a phage gene differs between the hosts, the correlation of expression between different hosts was compared to the correlation of expression between replicate time series for the same host. Changes in the expression of phage genes would be indicated if a low correlation of expression between hosts is observed in comparison to the correlation between replicated measurements. For statistical assessment, the average Pearson correlation coefficient between replicates was calculated for each gene individually. Then, the Fisher z-transformation was applied to the correlation coefficient to obtain an approximately normally distributed score, which can serve as empirical background distribution. Equally, correlation coefficients of the expression in different hosts were calculated for each phage gene and converted by Fisher z-transformation. To evaluate statistical significance, z-scores and the corresponding p-values were calculated for the later based on the mean and standard deviation of the empirical background distribution. Finally, p-values were adjusted for multiple testing using the Benjamini-Hochberg procedure.

Assessing the stability of phage gene clusters

The Jaccard coefficient was used to assess the stability of the number of clusters obtained from hierarchical clustering. The Jaccard coefficient provides a measure for the similarity of two different sets of clusters and ranges from zero (no similarity) to one (no difference). For statistical evaluation of the clustering stability, random subsets of the samples containing 70% of the genes were repeatedly (1000

times) selected and clustered and the Jaccard coefficient was calculated. This procedure was carried out with varying numbers of clusters and the distribution of the Jaccard coefficients obtained were displayed as histograms (Figure S6). High Jaccard coefficients signify high similarity between the original clusters and those obtained using subsets of the data, and are indicative of high stability for that particular number of clusters. The *benhur* function in the R/Bioconductor package, *clusterStab*, was used to carry out the clustering and calculation.

RT-PCR validation of microarray and RNA-seq results

The microarray and RNA-seq results were verified using quantitative RT-PCR (qRT-PCR) for 8 host and 2 phage genes during infection of the WH8102 host. These included host genes whose transcript levels increased immediately or subsequently as well as those for which the transcript levels remained unchanged or declined. The phage genes were from the early cluster. Transcript levels for each gene were normalized to host *rnpB* transcript levels. See Suppl. Table S1 for the genes tested and the primers and probes used.

Total RNA (10-180 ng) was reverse transcribed using the High Capacity cDNA Reverse Transcription Kit with RNase Inhibitor (Applied Biosystems) using 2 μ mol random primers in 20 μ l reactions following the manufacturer's instructions. The resultant cDNA was diluted 50 or 18 fold in 10 mM Tris pH8, and 10 μ l was used in each of 3 triplicate qPCR reactions using 1X Roche universal probe library (UPL) master mix (LightCycler® 480 Probes Master, Roche), 100 nM of appropriate UPL hydrolysis probe, 200 nM primers and template in a total volume of 25 μ l. As such, cDNA resulting from 0.1ng total RNA (for highly expressed phage and host genes) and 5 ng total RNA (for more lowly expressed host genes) was used in each qPCR reaction. These amounts were determined after ensuring representative transcripts were in the linear quantifiable range beginning from the RT step through to the qPCR analysis. Reactions were carried out on a LightCycler 480 Real-Time PCR System (Roche) (see above for the qPCR conditions). Results were further normalized to *rnpB* transcript levels which served as an internal control. No-RT controls, carried out under identical conditions but without the reverse transcriptase enzyme, indicated that gDNA contamination was less than 1 % of the RT-PCR signal in all samples. Standard curves were carried out with *Synechococcus* WH8102 or cyanophage Syn9 genomic DNA, as appropriate. Expression levels from infected cells at different times after infection was compared to that for control cells and are presented as fold change infected versus non-infected transcript levels.

Mass spectrometry identification of virion proteins

Phage particle preparation was carried out by polyethelene glycol precipitation followed by cesium chloride (CsCl) step gradients and dialysis as described previously (Lindell *et al.*, 2007) with modifications as described by Sabeji *et al.* (2012). Cesium chloride purified phage particles were denatured with 8 M urea and 100 mM ammonium bicarbonate, reduced with 2.8 mM DTT at 60 °C for 30 min, modified with 8.8 mM iodoacetamide for 30 min at 25 °C in the dark, and digested overnight at 37 °C with modified trypsin (Promega) at a 1:50 enzyme-to-substrate ratio in 2 M urea and 25 mM ammonium bicarbonate. Tryptic peptides were desalted and resuspended in 0.1% formic acid, resolved by reverse-phase chromatography on fused silica capillaries packed with Reprosil (Dr. Maisch GmbH), and eluted with 0.1% formic acid at a flow rate of 0.25 μ L·min⁻¹ with linear 95 min gradients of 7-40% acetonitrile and 8 min with 95% acetonitrile. Mass spectrometry (MS) was performed using an ion-trap mass spectrometer (Orbitrap, ThermoScientific) in positive mode with repetitive full MS scans followed by collision-induced dissociation of the five most dominant ions selected from the first scan. Data were analyzed using the Proteome Discoverer software (Thermo) searching against the *Synechococcus* sp. WH8102 section of the NCBI-nr database, against *Synechococcus* cyanophage Syn9 section of the NCBI-nr , and against decoy databases (in order to determine the false discovery rate). Proteomic analysis was carried out at the Technion Smoler Proteomics Center.

P-TIM40 cluster analysis

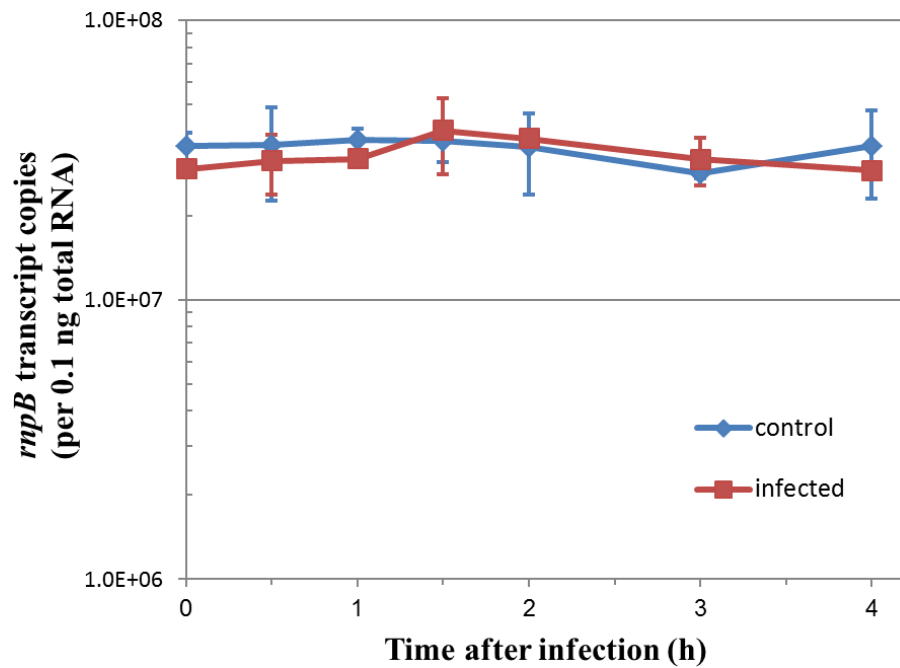
For clustering, RMA normalized expression values for the P-TIM40 phage were averaged over replicates and standardized, so that the expression of each gene had a mean value equal to 0 and a standard deviation equal to 1. These standardized values were “soft” clustered using the Bioconductor package *Mfuzz* (Futschik & Carlisle, 2005; Kumar & Futschik, 2007). Genes were clustered based on their $\log_2$ relative expression over time using the fuzzy c-means algorithm (FCM). The fuzzification parameter (m) was set to the default value for FCM, $m = 2$. The optimal number of clusters c ($c = 3$) was determined using Jaccard coefficient analysis in the same manner as for Syn9 following the procedure described in Lindell *et al.* (2007).

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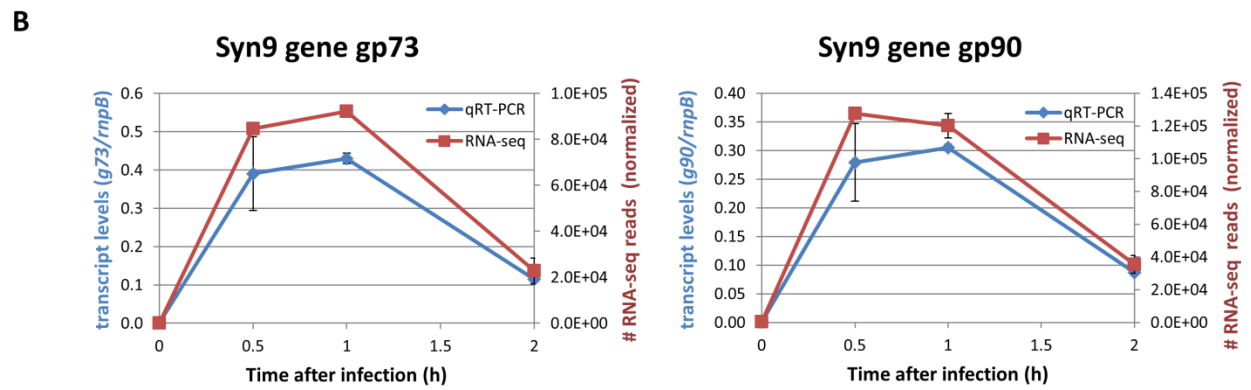
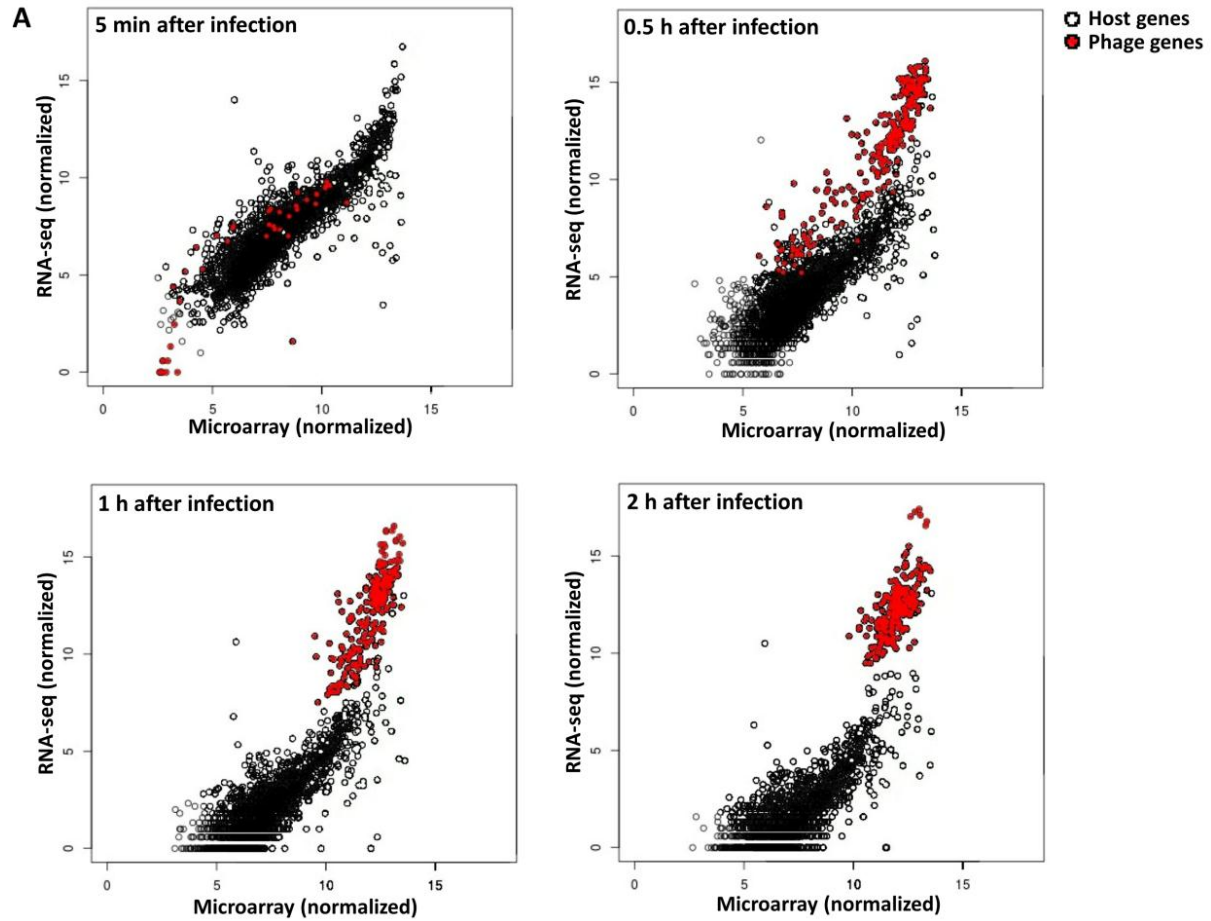
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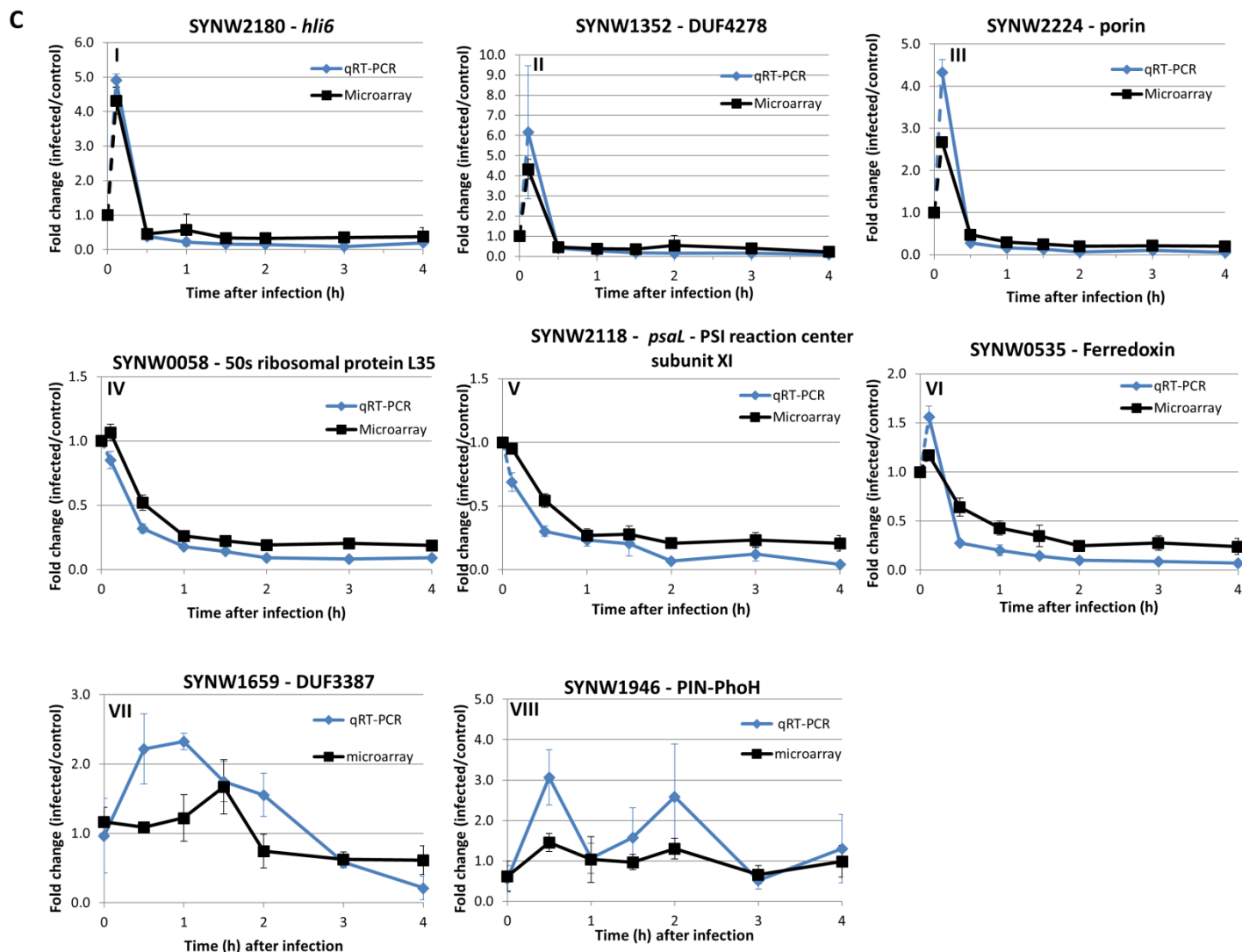
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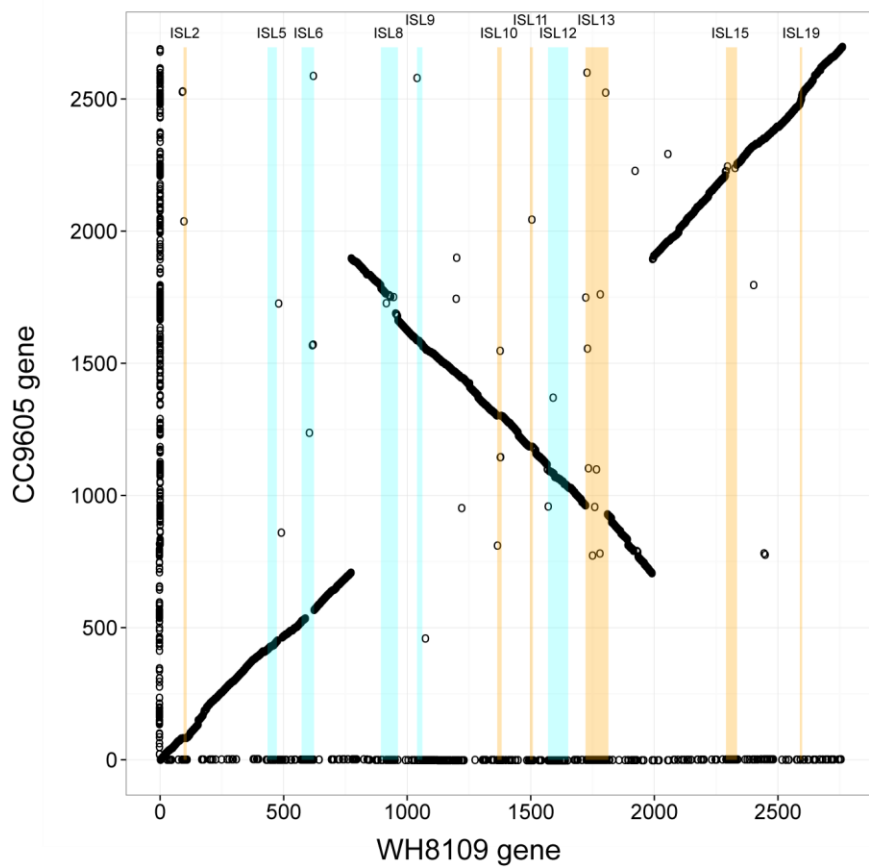


Supplementary Figure S1: Stable transcript levels of the *rnpB* gene during infection. Transcript levels of the *rnpB* gene from *Synechococcus* WH8102 during the 4 h latent period of infection by the Syn9 phage. Levels in infected cultures (shown in red) and uninfected control cultures (shown in blue) do not change over the course of infection, as determined by qRT-PCR. Results are normalized to the amount of total RNA used in each reaction and are shown in log scale. Average and range of 2 biological replicates. Three technical replicates were carried out for each biological replicate.

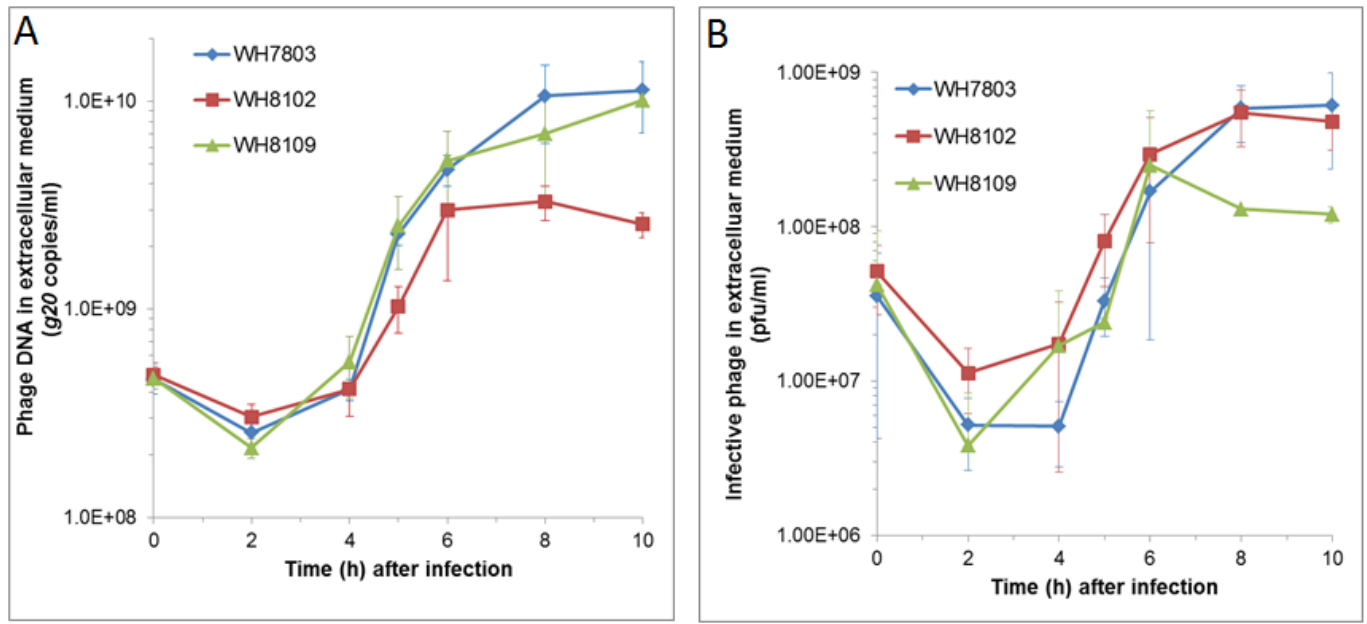




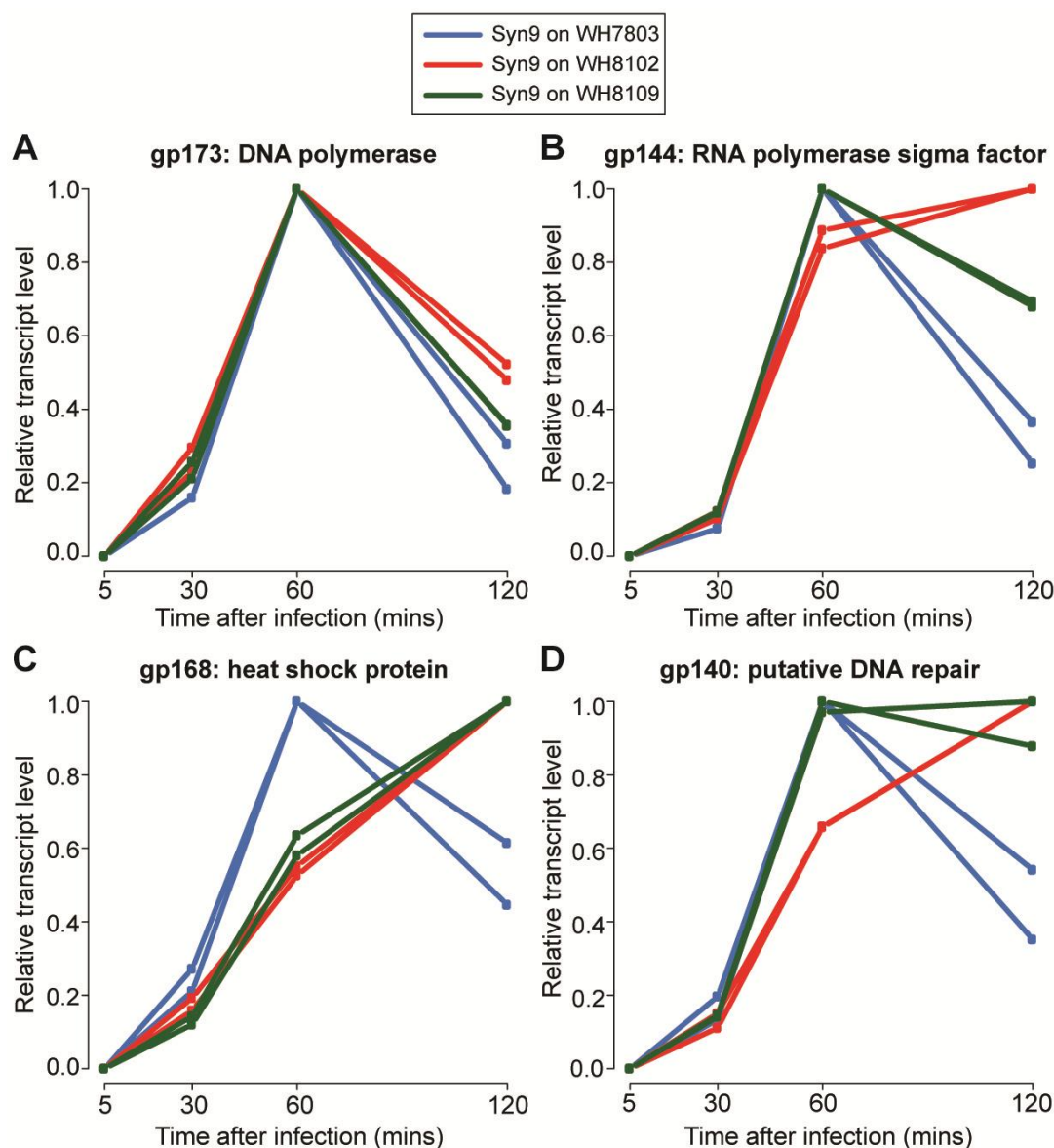
Supplementary Figure S2: Comparison of microarray and RNA-seq measurements. (A) Comparison of the expression levels measured by RNA-seq and microarray analyses. RNA-seq data were normalized to gene length and library size (RPKM) and the microarray data were RMA normalized. The replicates were averaged and are presented as log₂. Phage and host genes are shown with red and black circles, respectively. (B, C) Verification, using qRT-PCR, that phage and host transcript profiles are well-represented from RNA-seq (B) and microarray (C) analyses, respectively. Examples of early phage gene expression determined by qRT-PCR and RNA-seq analysis (B). Examples of host gene expression determined by qRT-PCR and microarray analysis (C) of immediately increased genes (I-III), subsequently declined genes (IV-VI), subsequently increased gene (VII) and a gene with unchanged transcript levels (VIII). Dashed lines indicate the change in transcript levels in the uninfected control cultures prior to infection relative to the infected cultures at 5 min after infection. Error bars represent the average and standard deviation of 3 biological replicates in the microarray data, and 2 biological replicates in the RNA-seq and qRT-PCR data (except for panels VII and VIII which represent 3 biological replicates in both methods).



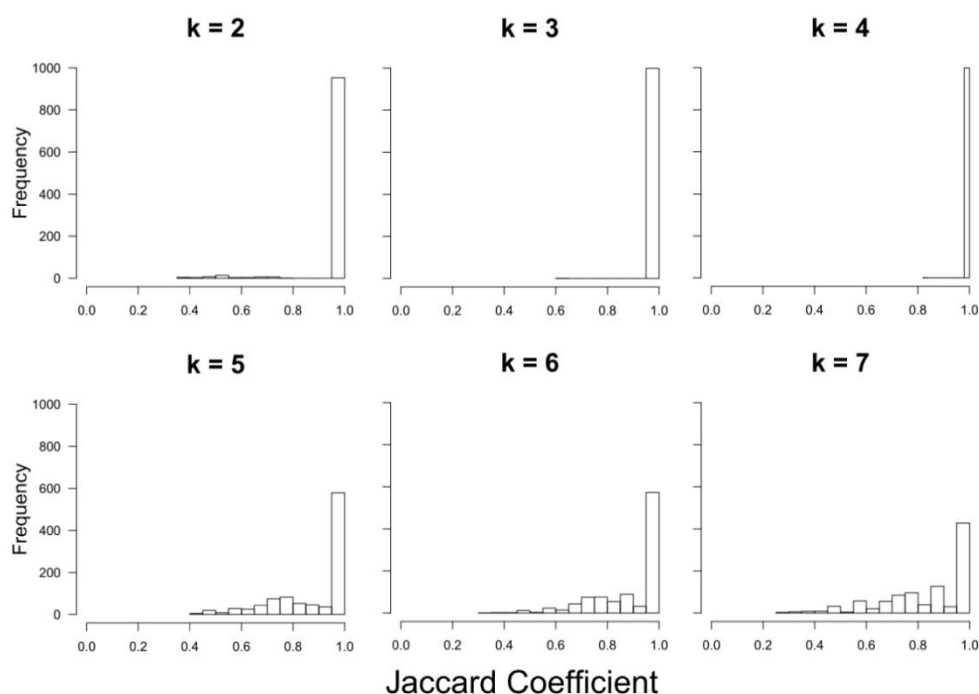
Supplementary Figure S3: Genome alignment of *Synechococcus* strains WH8109 and CC9605. Orthologous genes are shown in circles; strain-specific genes are on the axes. The locations of the major islands are shaded (orange for islands that are unique to *Synechococcus* WH8109 and blue for islands that are common to *Synechococcus* CC9605). See Suppl. Table S13 for details of the location and composition of the genomic islands *Synechococcus* WH8109.



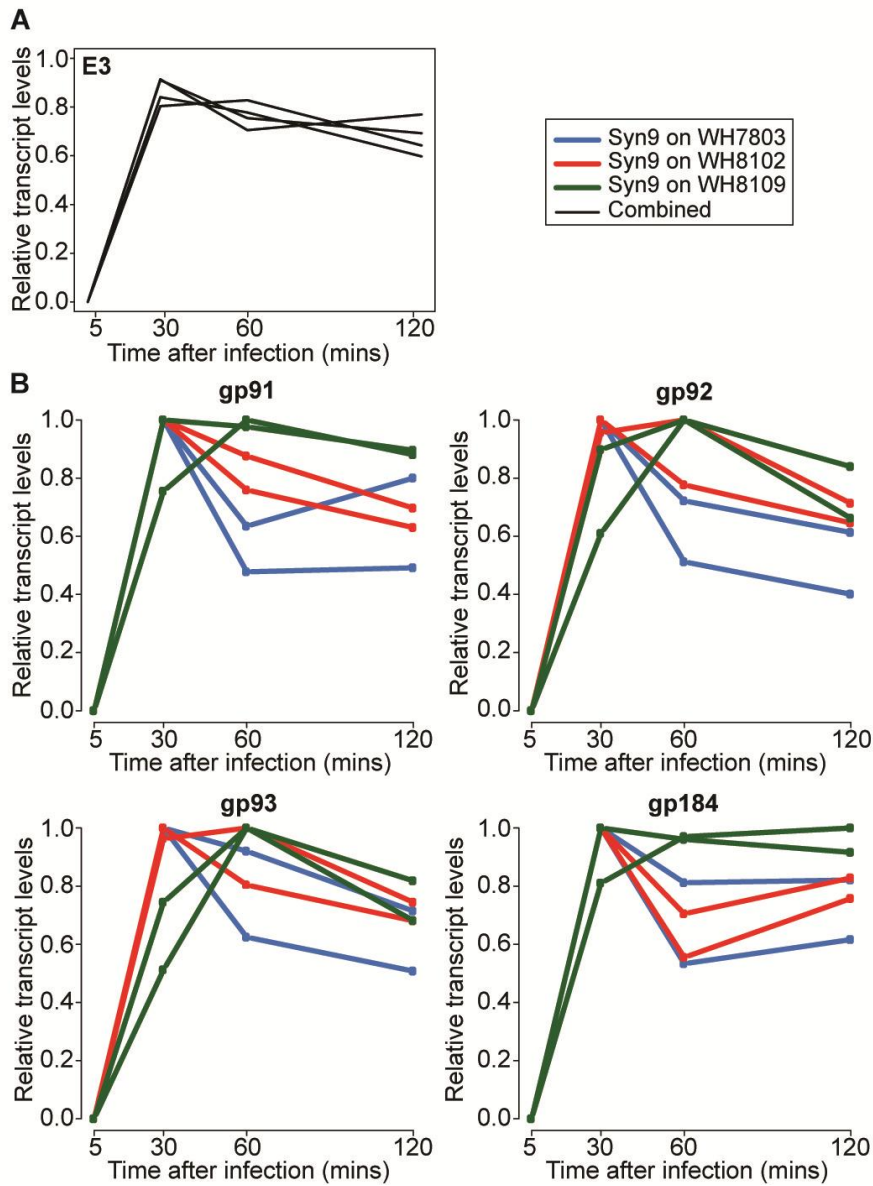
Supplementary Figure S4: Comparison of phage DNA to infective phage yields. (A) Quantification of phage DNA in the extracellular medium determined by qPCR of the Syn9 g20 gene. (B) Number of infective phage released to the extracellular medium as determined by the plaque assay. (n=3 for all measurements except infective phage on WH8109 where n=2). Note that extracellular DNA is made up of both free DNA released from the cell and encapsidated DNA in both infective and non-infective phage particles. Panel B is the same as Figure 1B and is shown here for comparative purposes only.



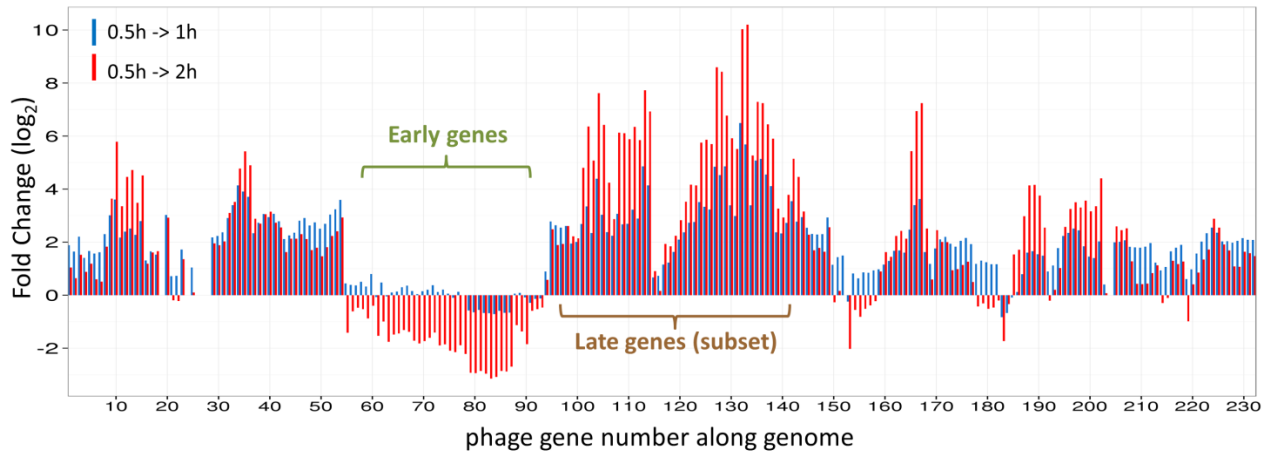
Supplementary Figure S5: Reproducibility and divergence of phage gene expression patterns in different hosts. Expression patterns of phage genes during infection of the different *Synechococcus* hosts, WH7803, WH8102 and WH8109, are shown with blue, red, and green lines, respectively. Each biological replicate is shown separately. The normalized number of reads was adjusted to minimum (0) and maximum (1) values, represented by the Y-axis. (A) The DNA polymerase gene (gp173) is an example of reproducible expression in the 3 hosts and is representative of the overwhelming majority of genes. (B-D) Genes with seemingly differential expression patterns on the different hosts. Note that differential expression patterns were not statistically significant for any of the genes (see Suppl. Methods, under “Comparing phage transcription patterns between the different hosts”).



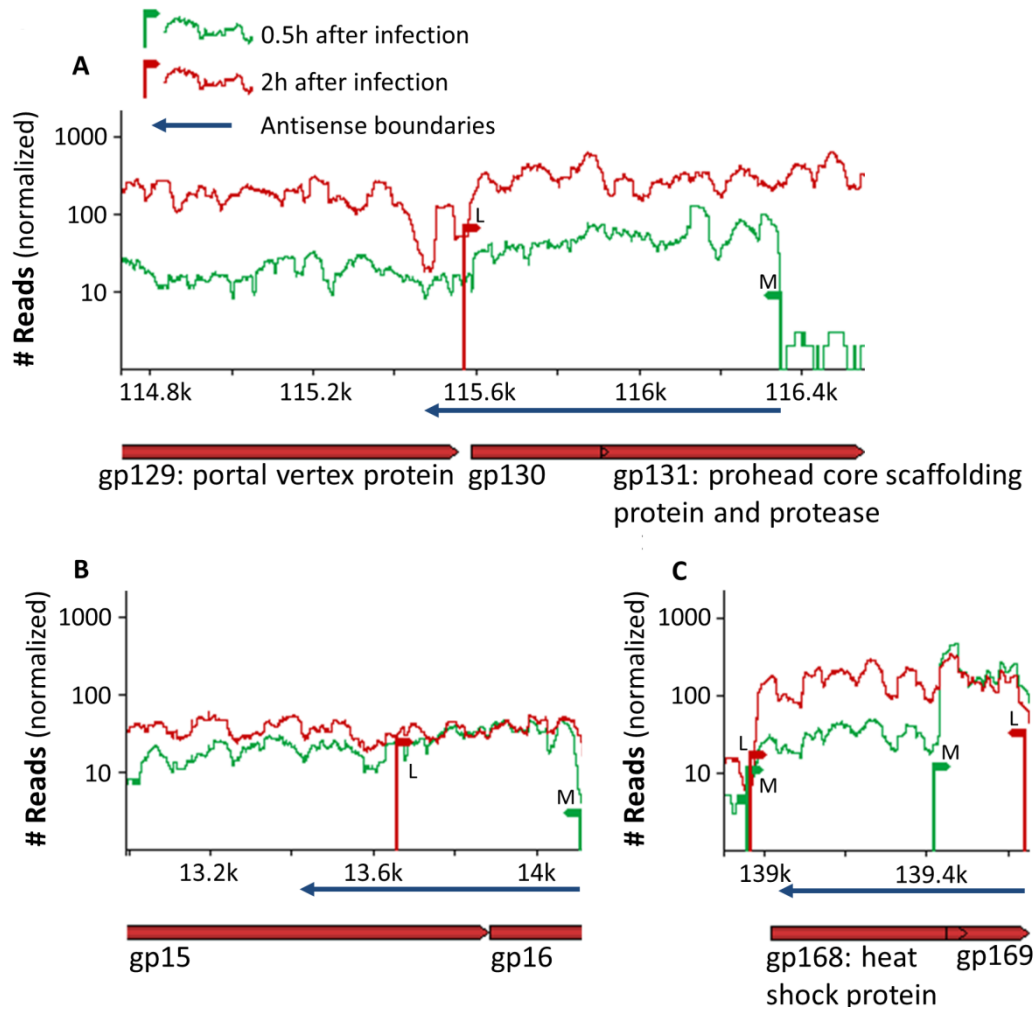
Supplementary Figure S6: Jaccard similarity coefficients. The Jaccard coefficient was used as a stability measure in order to determine the most stable number of temporal classes of transcription (clusters) of the phage genes. Individual histograms show the distribution of the Jaccard coefficients for 1000 subsets of genes when 70% of the genes are randomly selected for each subset. Each histogram displays the distribution of Jaccard coefficients for a different number of clusters, k ($k = 2$ to $k = 7$). Clustering is considered stable when the majority of Jaccard values are close or equal to one. In our case the histograms for $k = 3$ and $k = 4$ show that the vast majority of Jaccard coefficients are close to one indicating that they are the most stable clustering solutions. See Suppl. Text for additional details.



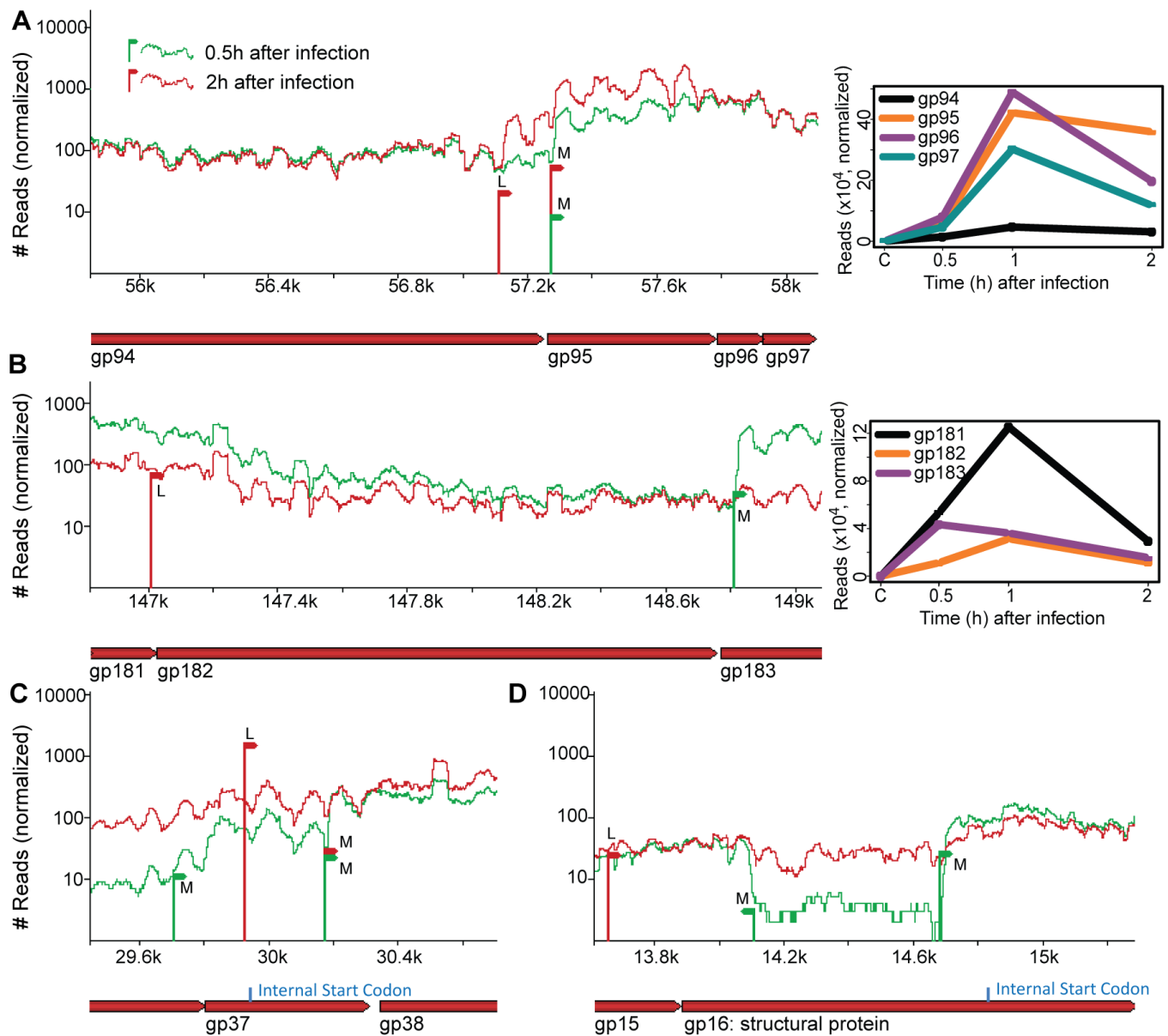
Supplementary Figure S7: Transcript levels of the four phage genes in the third early cluster, E3. (A) Expression profiles of the genes in this subcluster as a function of time after infection. Relative transcript levels were normalized to minimum (0) and maximum (1) values for each gene and were averaged across all 3 hosts (combined). (B) Expression profiles of the individual phage genes of this subcluster during infection of the different hosts, *Synechococcus* strains WH7803, WH8102 and WH8109 shown with blue, red, and green lines, respectively. Each biological replicate is shown separately. Relative transcript levels were normalized to minimum (0) and maximum (1). We note that the four genes in this cluster showed uncharacteristically divergent patterns between replicates and hosts.



Supplementary Figure S8: Genome organization of phage early, middle and late genes. Fold change of phage gene expression during infection of the *Synechococcus* WH8102 host, between 0.5 h and 1 h (blue) and between 0.5 h and 2 h (red) after infection. The early and late genes with the highest fold changes are indicated on the graph.

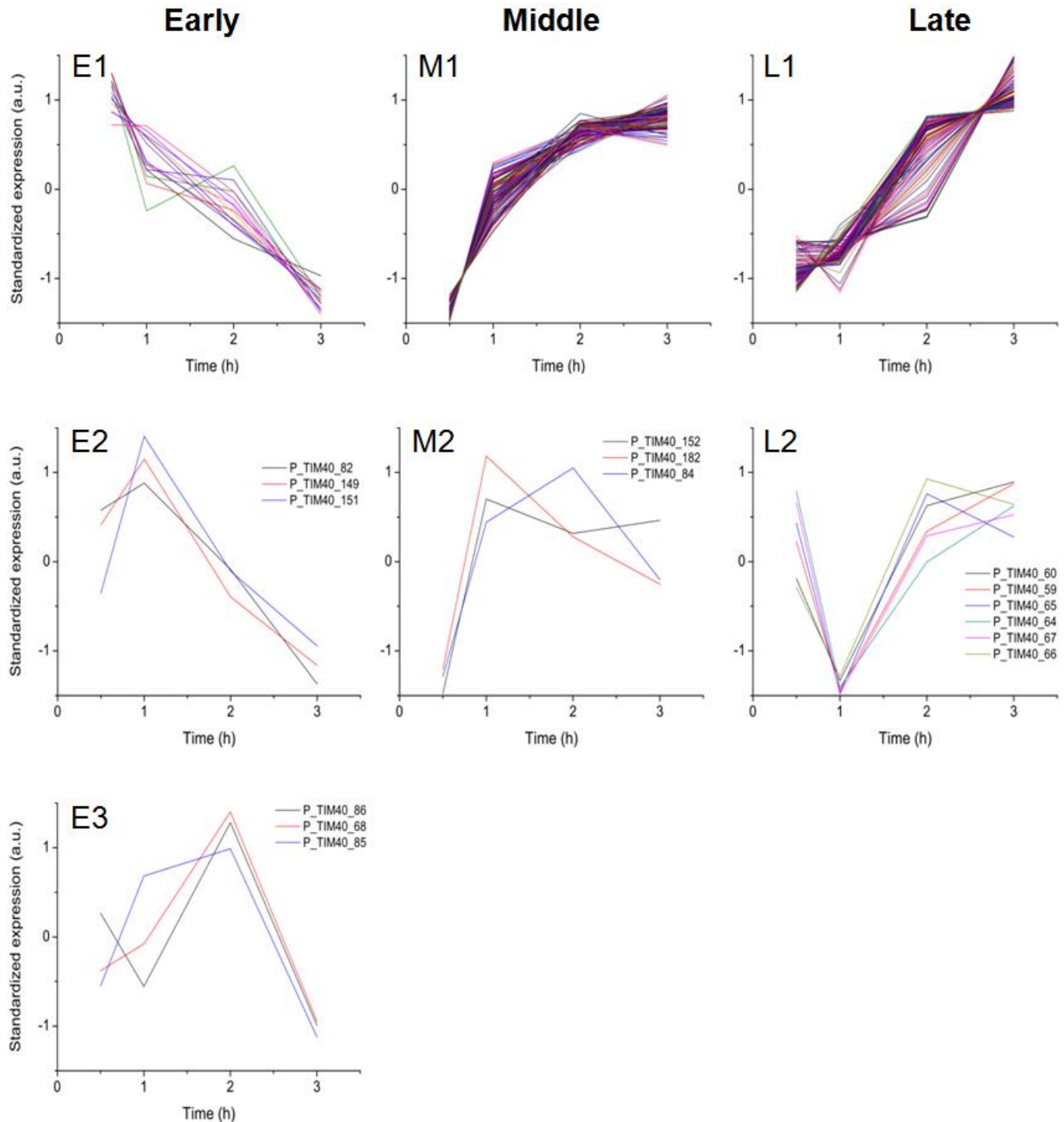


Supplementary Figure S9: Temporal expression of antisense RNAs (asRNA) during infection. (A) An asRNA to g131 and g130 is transcribed from a middle promoter 0.5 h after infection, while at 2 h after infection mRNA of g130 and g131 is transcribed from a late promoter. (B) An asRNA to g16 and g15 is transcribed from a middle promoter 0.5 h after infection, while at 2 h after infection mRNA of g16 is transcribed from a late promoter. (C) An asRNA to g169 and g168 is transcribed from a late promoter at 2 h after infection, while mRNA of g168 and g169 is transcribed from a middle promoter at 0.5 h after infection as well as from a late promoter at 2 h after infection. X-axis, position on the phage genome; phage gene products are represented by red block arrows below the X-axis. The RNA-seq reads (Y-axis) were normalized to library size. The letters E, M and L on the TSS vertical lines indicate whether the associated promoter is an early, middle or late promoter, respectively. Antisense transcript lengths (inferred from bisulfite sequencing) are shown by the blue arrows (see Suppl. Fig. S15).



Supplementary Figure S10: Alternative promoters generate complex temporal transcription of phage genes.

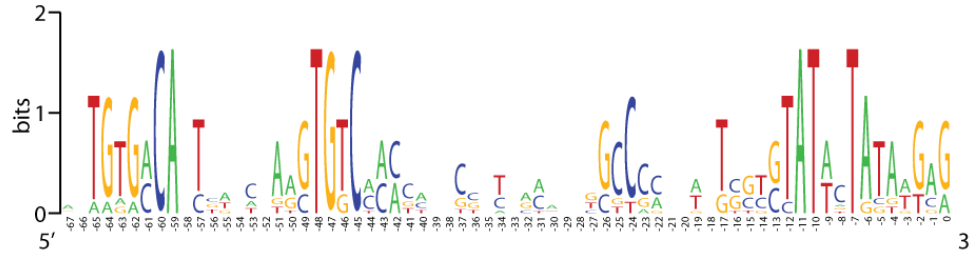
(A) Example of a gene regulated by different promoters at different time points. Gene 95 is transcribed from both a middle and a late promoter. (B) Example of modular operon structure where g183 is transcribed from its own middle promoter 0.5 h after infection, and from an upstream late promoter 2 h after infection as a part of the g182 operon. (C, D) Examples of modular genes: (C) A late promoter internal to g37 may produce a short subprotein of 122 amino acids, while the full-length g37 gene, transcribed from a middle promoter, would produce a protein of 168 amino acids. (D) A middle promoter internal to g16 may produce a short subprotein of 153 amino acids, while the full-length gp16 gene transcribed from a late promoter would produce a protein of 467 amino acids. Expression profiles of the relevant genes are shown to the right. X-axis, position on the phage genome; phage gene products are represented by red block arrows below the X-axis. The letters E, M and L on the TSS vertical lines indicate whether the associated promoter is an early, middle or late promoter, respectively. The blue lines in panels (C) and (D) indicate internal start codons.



Supplementary Figure S11. The temporal expression program of the P-TIM40 phage. Expression profiles of individual P-TIM40 phage genes in each expression cluster: early, middle and late transcription. The profiles are separated into subclusters based on similarity of transcriptional patterns of the individual genes. The cluster and subcluster designations are shown in the top left corner of each graph. Membership of each of the genes is shown in Suppl. Table S8. Relative transcript levels (Y-axes) were averaged over the 3 biological replicates and standardized so that the expression of each gene has a mean value of 0 and a standard deviation of 1.

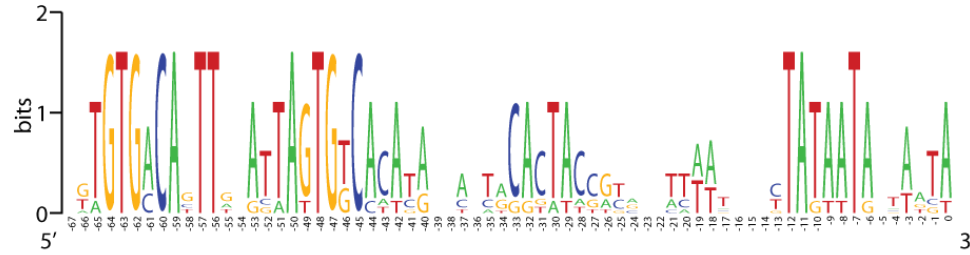
Syn9

9 promoters



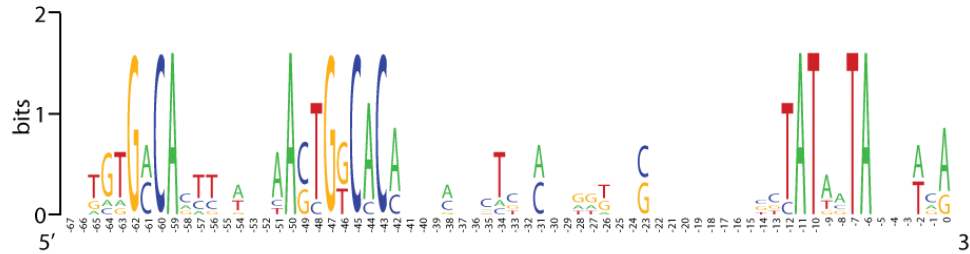
P-TIM40

8 promoters



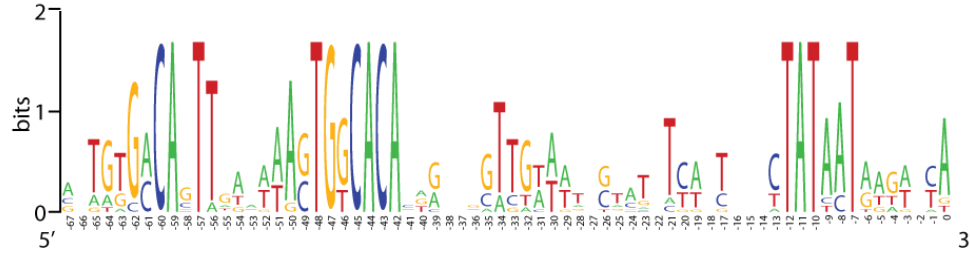
S-PM2

8 promoters



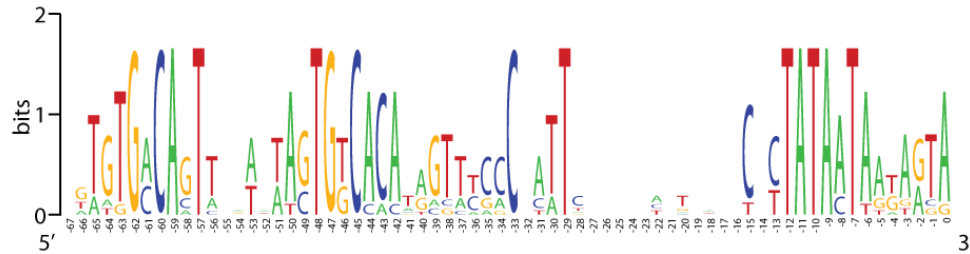
S-SSM7

12 promoters



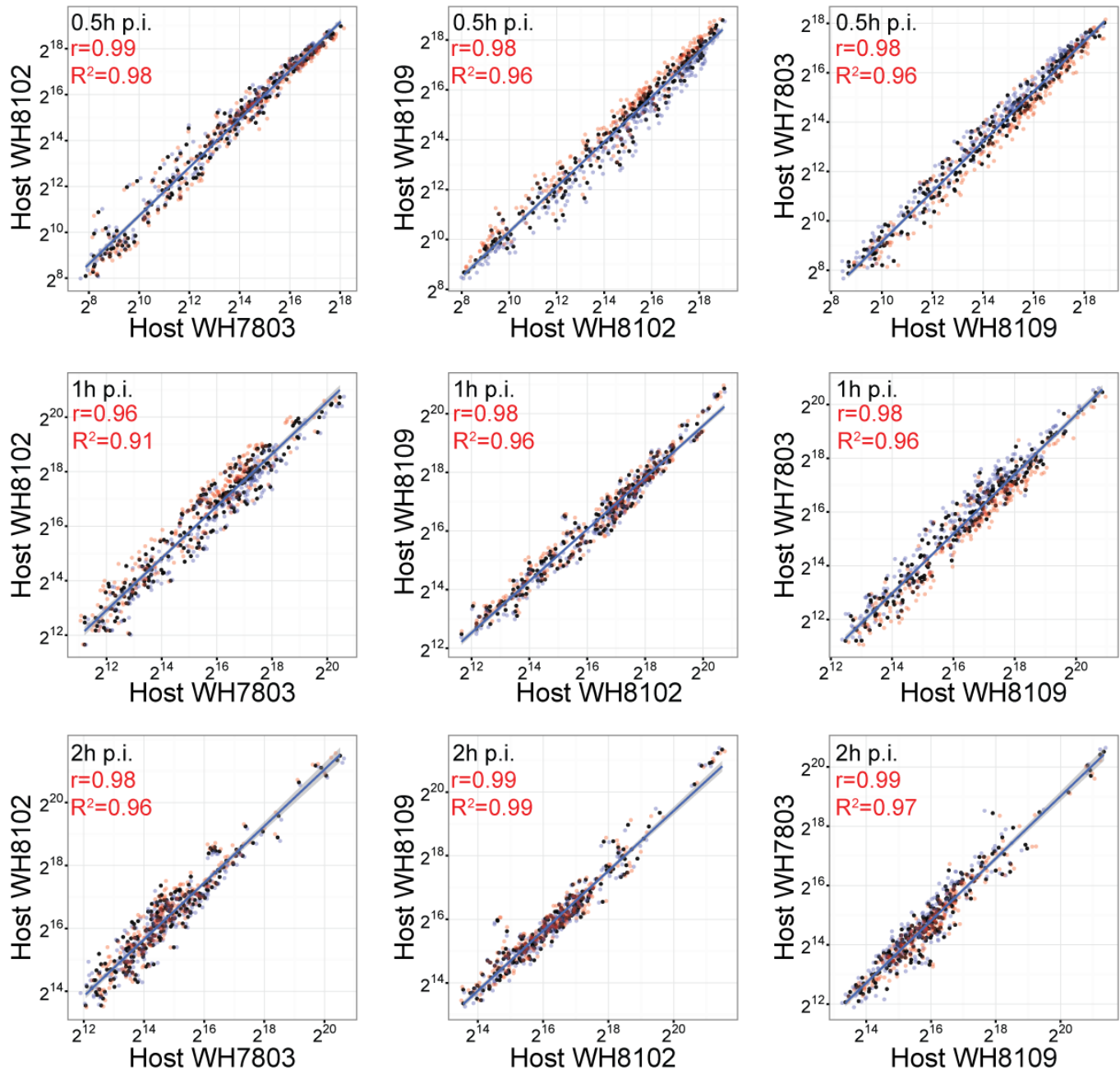
P-RSM4

10 promoters

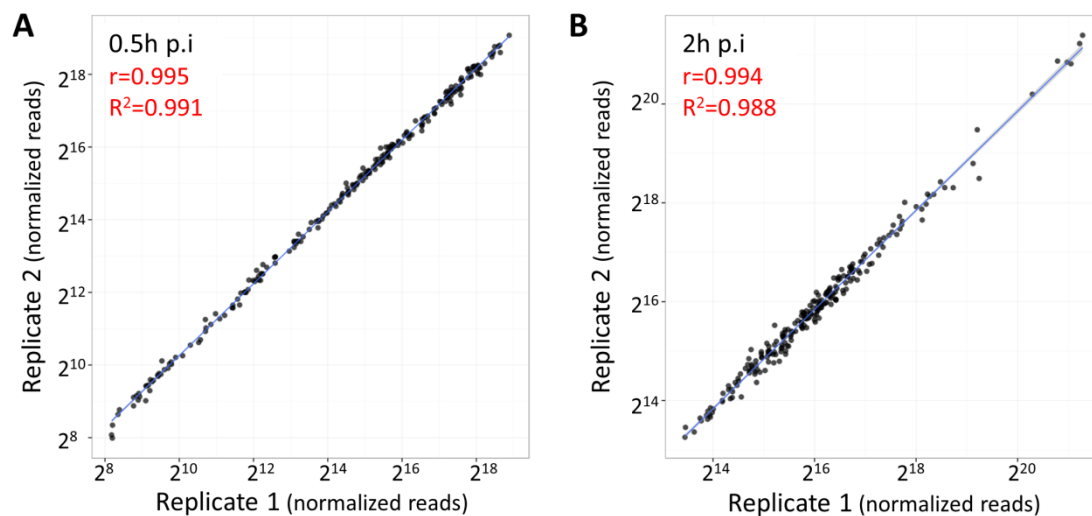


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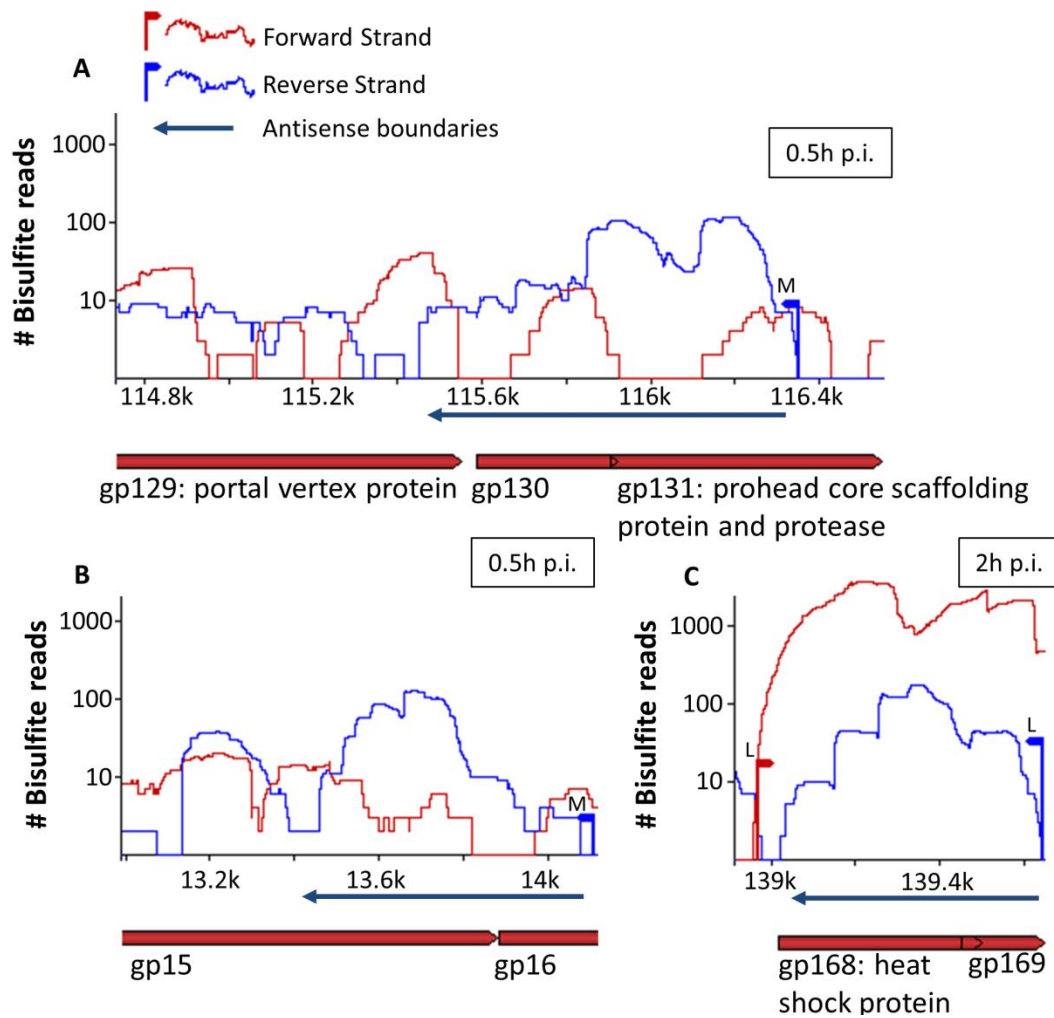
Supplementary Figure S12: Conservation of the early promoter motif in T4-like cyanophages. Early promoter motifs in P-TIM40 and in representatives of the three major T4-like cyanomyovirus clades are presented (S-PM2 belongs to clade I, S-SSM7 belongs to clade II and P-RSM4 belong to clade III, following Sullivan *et al.* (2008)). The motif was also detected in the following phages: P-SSM2, P-SSM4, S-SM1, P-HM1, P-HM2, Syn19, S-SSM5, P-SSM7, Syn33, S-ShM2 and Syn1. The Syn9 motif presented here is that determined experimentally in this study and is displayed here for comparative purposes. Promoter sequences were aligned prior to generating the logo.



Supplementary Figure S13: High correlation in phage gene expression across hosts. Correlation of phage gene expression at the different time points post infection (p.i.), between all of the hosts. Expression of phage genes averaged over two biological replicates is shown in black and biological replicates are shown in red and blue (normalized to host *rnpB* transcript levels). Pearson correlation coefficients using linear values (r) and coefficients of determination (R^2) are shown in the respective panels. Pearson correlation coefficients using logged values were between 0.95 and 0.99.



Supplementary Figure S14: Reproducibility between biological replicates. Correlation of phage gene expression between two biological replicates, (A) at 0.5 h after infection of the *Synechococcus* WH8102 host, and (B) at 2 h after infection of the *Synechococcus* WH8109 host. The Pearson correlation coefficients (r) using linear values are shown in each panel and both had p -values $< 2.2e-16$. Coefficients of determination (R^2) are also shown in each panel. Pearson correlation coefficients using logged values were 0.999 and 0.994 for (A) and (B), respectively.



Supplementary Figure S15: Estimating asRNA length from bisulfite sequencing. Phage gene expression from the forward strand (red) and the reverse strand (blue) after infection of the *Synechococcus* WH8109 host, as shown from strand-specific bisulfite sequencing. X-axis, position on the phage genome; phage gene products are represented by red block arrows below the X-axis. Y-axis, the number of bisulfite RNA-seq reads. Antisense transcript lengths were estimated based on the sharp decline in coverage of bisulfite sequencing reads, and are shown by the blue arrows. Promoter positions are designated with vertical lines and were determined by 5'-sequencing. The letters M and L indicate whether the associated promoter is a middle or late promoter, respectively. (A, B, C) Examples of asRNAs are those shown in Suppl. Fig. S9.

Supplementary Table S1: Primers used for qPCR and qRT-PCR

Target gene and product	UPL probe no.	Primer name	Primer sequence 5'-3'
Phage Syn9 genes			
BSV9_gp129 <i>g20</i> - portal protein	84	Syn9_gp20_UPL84_F	TCGTTTAGAAACAGAAACCACATTT
		Syn9_gp20_UPL84_R	AACTTTGGGAATTTAACTTCGTCAC
BSV9_gp73 hypothetical protein	145	Syn9_gp90_UPL145_F	TGGGTATGATTGATGCCAAG
		Syn9_gp90_UPL145_R	TAACCTCAGCACCGTTAGCA
BSV9_gp90 hypothetical protein	113	Syn9_gp73_UPL113_F	TCGCTAACAGCAAAGACTGG
		Syn9_gp73_UPL113_R	GAGGTGTTGTTTTCTTCGTTGA
Host WH8102 genes			
SYNW1718 <i>rbcL</i> - Rubisco large subunit	85	rbcL_SynsCons-F	TTCATCAAGAGCTGCTACGG
		rbcL_SynsCons-R	GACGGCCGTACTTGTTTCATC
RNA_3 <i>rnpB</i> - RNA component of RNaseP	91	RNPB_Fw_UPL91	CATCGGCGGTGTGTTTCT
		RNPB_Rv_UPL91	CAGGCTTGCTGGGTAACG
SYNW0058 <i>rpmI</i> - 50S ribosomal protein L35	62	SYNW0058 #62 Fw	TTCTGCTTGGGGGTCTTGT
		SYNW0058 #62 Rv	CAAGCGTTCAAGGCAAC
SYNW0535 <i>petF4</i> - ferredoxin	146	SYNW0535 #146 Fw	AGTGCCCGACGACCAGTA
		SYNW0535 #146 Rv	GAGTAGGGCAGGTCGATGC
SYNW1352 hypothetical protein	138	SYNW1352#138 Fw	CAGAAGCCACAACCTGCGTTA
		SYNW1352#138 Rv	ATAGCTCACGCCGCGATA
SYNW1659 conserved hypothetical protein [similar to C-terminal end of type I restriction enzyme HsdR]	13	SYNW1659#13 Fw	CCGCCTCCCCTATAAGGTC
		SYNW1659#13 Rv	CGCCATTGATCTTGTGCTC
SYNW1946 PhoH family protein	131	SYNW1946#131Fw	CGCTCAATTCCCCAACAG
		SYNW1946#131Rv	CACTTCCAAGGGCGTCAG
SYNW2106 hypothetical protein	161	SYNW2106#161Fw	GGTTTCAGCGTCCTCCTG
		SYNW2106#161Rv	GGAATGGCAATGCCTGAC
SYNW2118 <i>psaL</i> - photosystem I subunit XI	62	SYNW2118 #62 Fw	TATACCGTCCCGGCATCTC
		SYNW2118 #62 Rv	GGACCGCAGATGGTGAAG
SYNW2180 <i>hli6</i> - high light inducible protein	146	SYNW2180#146Fw	CATGATGGGCTTCGTCATC
		SYNW2180#146Rv	CTGGGACAGGATGCCTTG
SYNW2224 <i>som</i> - possible porin	62	SYNW2224#62Fw	AGCGCTGATTGAGGGAATC
		SYNW2224#62Rv	CTTTGCTTTCAACGGCTTCT

Supplementary Table S2: Summary of sequencing data – Syn9 phage infecting host WH7803

Time point	5 min		0.5 h		1 h		2 h	
Replicate	1	2	1	2	1	2	1	2
Protocol	Whole-transcriptome sequencing							
Number of reads	28,340,539	13,924,181	14,586,185	15,805,517	26,942,514	10,872,247	13,473,297	11,909,927
Mapped reads	26,202,060	12,851,218	12,482,206	15,130,410	23,537,945	10,533,721	11,899,731	11,549,224
% mapped reads	92.5%	92.3%	85.6%	95.7%	87.4%	96.9%	88.3%	97.0%
% mapped to non-rRNA regions	8.7%	10.2%	29.7%	33.5%	59.7%	60.7%	45.2%	51.1%
Mapped to phage	1,917	79	3,265,317	4,641,509	13,718,800	6,290,369	5,229,496	5,802,699
Mapped to host (non-rRNA regions)	2,285,895	1,314,738	446,093	425,866	329,283	99,482	147,945	96,399
% mapped to phage / mapped to non-rRNA	0.1%	0.0%	88.0%	91.6%	97.7%	98.4%	97.2%	98.4%
% mapped to host / mapped to non-rRNA	99.9%	100.0%	12.0%	8.4%	2.3%	1.6%	2.8%	1.6%
Protocol	5'-end sequencing							
Number of reads			2,329,227	14,010,049	3,040,942	12,065,225	1,649,751	6,914,762
Mapped reads			1,684,197	7,534,757	1,244,305	3,137,525	951,679	3,173,695
% mapped reads			72.3%	53.8%	40.9%	26.0%	57.7%	45.9%
Mapped to phage			166,497	1,065,152	133,245	560,202	359,904	873,997

Supplementary Table S3: Summary of sequencing data – Syn9 phage infecting host WH8102

Time point	5 min		0.5 h		1 h		2 h	
Replicate	1	2	1	2	1	2	1	2
Protocol	Whole-transcriptome sequencing							
Number of reads	22,595,081	21,897,362	14,042,556	21,497,710	14,656,919	25,226,041	11,903,607	13,873,290
Mapped reads	22,391,866	21,816,462	12,723,542	21,358,476	13,434,541	25,098,252	10,929,163	13,799,948
% mapped reads	99.1%	99.6%	90.6%	99.4%	91.7%	99.5%	91.8%	99.5%
% mapped to non-rRNA regions	12.4%	13.1%	31.0%	37.8%	56.1%	53.0%	51.5%	51.5%
Mapped to phage	4,048	7,145	3,338,520	7,235,442	7,297,362	12,867,340	5,500,831	6,944,354
Mapped to host (non-rRNA regions)	2,762,262	2,842,856	606,222	827,796	236,112	422,369	130,921	163,005
% mapped to phage / mapped to non-rRNA	0.1%	0.3%	84.6%	89.7%	96.9%	96.8%	97.7%	97.7%
% mapped to host / mapped to non-rRNA	99.9%	99.7%	15.4%	10.3%	3.1%	3.2%	2.3%	2.3%
Protocol	5'-end sequencing							
Number of reads			11,876,902	5,085,865	13,968,128	4,159,581	3,014,467	15,546,034
Mapped reads			3,557,109	2,653,953	2,771,958	1,523,097	1,711,598	7,821,337
% mapped reads			30%	52%	20%	37%	57%	50%
Mapped to phage			145,713	294,132	429,470	126,788	439,807	1,956,214

Supplementary Table S4: Summary of sequencing data – Syn9 phage infecting host WH8109

Time point	5 min		0.5 h		1 h		2 h	
Replicate	1	2	1	2	1	2	1	2
Protocol	Whole-transcriptome sequencing							
Number of reads	19,176,315	54,656,158	56,158,341	19,247,098	57,797,114	5,868,532	42,366,058	37,929,692
Mapped reads	19,108,538	54,340,430	55,845,098	19,173,724	57,578,252	5,849,888	42,183,875	37,796,009
% mapped reads	99.7%	99.4%	99.4%	99.6%	99.6%	99.7%	99.6%	99.7%
% mapped to non-rRNA regions	15.4%	15.8%	20.7%	14.7%	39.5%	37.4%	36.1%	32.1%
Mapped to phage	38	216	9,654,908	2,248,083	21,897,872	2,103,634	14,813,759	11,776,814
Mapped to host (non-rRNA regions)	2,952,215	8,591,383	1,909,616	578,792	828,491	85,215	393,823	362,325
% mapped to phage / mapped to non-rRNA	0.0%	0.0%	83.5%	79.5%	96.4%	96.1%	97.4%	97.0%
% mapped to host / mapped to non-rRNA	100.0%	100.0%	16.5%	20.5%	3.6%	3.9%	2.6%	3.0%
Protocol	5'-end sequencing							
Number of reads			57,204,907	4,248,684	40,988,707	5,433,021	54,713,675	7,812,126
Mapped reads			27,108,240	3,892,224	22,483,135	4,588,191	14,364,676	5,587,416
% mapped reads			47.4%	91.6%	54.9%	84.5%	26.3%	71.5%
Mapped to phage			3,022,436	617,090	5,250,029	2,245,169	3,561,218	1,987,662

Supplementary Table S6: Syn9 virion particle proteins detected by mass spectrometry.

	locus_tag	description	MW [kDa]	Coverage (%)	No. of Unique Peptides
1	BSV9_gp8	baseplate wedge subunit (t4-gp53)	32.9	31	7
2	BSV9_gp9	putative baseplate tail tube cap (t4-gp48)	36.8	30	7
3	BSV9_gp11	conserved hypothetical protein	32.2	5	2
4	BSV9_gp14	conserved hypothetical protein	118.8	11	10
5	BSV9_gp15	conserved hypothetical protein	110.1	27	23
6	BSV9_gp16	structural protein	53.4	28	12
7	BSV9_gp20	putative baseplate hub subunit (t4-gp5)	87.1	4	3
8	BSV9_gp32	structural protein (putative fibre)	48.0	55	24
9	BSV9_gp33	structural protein (putative fibre)	51.7	38	14
10	BSV9_gp34	structural protein (putative fibre)	142.1	32	41
11	BSV9_gp35	conserved hypothetical protein	32.3	14	5
12	BSV9_gp94	conserved hypothetical protein	107.3	6	6
13	BSV9_gp99	baseplate wedge subunit (t4-gp25)	15.7	43	6
14	BSV9_gp100	baseplate wedge subunit & tail pin, (t4-gp6)	74.0	40	27
15	BSV9_gp101	structural protein (putative fibre)	655.3	25	117
16	BSV9_gp102	baseplate wedge subunit and tail pin, (t4-gp10)	42.2	23	5
17	BSV9_gp103	hypothetical protein (putative fibre)	151.0	22	22
18	BSV9_gp104	baseplate wedge (t4-gp8)	55.1	42	19
19	BSV9_gp105	structural protein	262.2	39	82
20	BSV9_gp108	structural protein (putative fibre)	183.6	32	38
21	BSV9_gp109	neck protein (t4-gp13)	31.0	25	6
22	BSV9_gp110	neck protein (t4-gp14)	42.3	24	9
23	BSV9_gp111	tail sheath stabilizer (t4-gp15)	29.9	35	9
24	BSV9_gp114	conserved hypothetical protein	49.8	19	9
25	BSV9_gp127	tail sheath (t4-gp18)	81.0	42	37
26	BSV9_gp128	tail tube (t4-gp19)	22.6	58	10
27	BSV9_gp129	portal (t4-gp20)	61.4	40	24
28	BSV9_gp131	prohead core scaffolding protein & protease, (t4-gp21)	23.9	19	3
29	BSV9_gp132	scaffold (t4-gp22)	38.0	19	6
30	BSV9_gp133	major capsid (t4-gp23)	48.7	44	26
31	BSV9_gp138	tail completion and sheath stabilizer, (t4-gp3)	21.8	15	3
32	BSV9_gp182	conserved hypothetical protein	64.4	22	10
33	BSV9_gp184	conserved hypothetical protein	42.6	50	16
34	BSV9_gp185	conserved hypothetical protein	57.1	23	10
35	BSV9_gp186	conserved hypothetical protein	27.6	7	2
36	BSV9_gp188	structural protein	20.2	58	8
37	BSV9_gp190	conserved hypothetical protein	19.8	34	7
38	BSV9_gp220	conserved hypothetical protein	127.6	10	9

Supplementary Table S9: Host genes with an immediate increase in transcript levels. Log₂ fold change (FC) of infected versus control cultures with time (h) after infection. Positive and negative values indicate an increase and decrease in transcript levels during infection with significant differences in Log₂ FC (q<0.05) shown in red and blue, respectively.

Strain/Host	Functional group	Gene	Predicted function	Log ₂ FC (infected/control)						
				0h	0.5h	1h	1.5h	2h	3h	4h
WH8102	Cell envelope *	SYNW1633	putative <i>murA</i> : UDP-N-glucosamine 1-carboxyvinyltransferase	0.74	-1.32	-1.85	-1.65	-1.78	-1.82	-2.42
		SYNW2187	putative RND family outer membrane efflux protein	1.26	-1.25	-1.50	-1.66	-1.83	-1.97	-2.69
		SYNW2224 ^a	porin (P-stress induced)	1.46	-1.07	-1.73	-2.02	-2.31	-2.22	-2.32
		SYNW2226	<i>som</i> : possible porin-likely pseudogene	1.29	-1.17	-1.74	-1.85	-2.36	-2.21	-2.58
	Carbon fixation *	SYNW1712 ^b §	<i>csoS1/ccmK2</i> : possible CO ₂ concentrating mechanism protein	0.65	-1.12	-2.10	-2.16	-2.41	-2.26	-2.43
		SYNW1713	putative carboxysome peptide B	0.94	-1.08	-2.18	-2.13	-2.26	-2.12	-2.52
		SYNW1714	putative carboxysome peptide A	0.76	-1.15	-1.91	-2.04	-2.35	-2.10	-2.33
		SYNW1715	<i>csoS3</i> : putative Carboxysome shell polypeptide	1.09	-1.13	-1.84	-1.82	-2.00	-2.01	-2.31
	High light inducible stress	SYNW0053 §	similar to high light-inducible protein	1.07	-1.46	-1.78	-1.95	-2.07	-2.29	-2.51
		SYNW2180	<i>hli6</i> : possible high light inducible protein	2.06	-1.15	-1.10	-1.61	-1.60	-1.49	-1.61
	DNA repair	SYNW2266	Putative sugar-phosphate nucleotidyl transferase	0.57	-1.39	-1.95	-2.00	-2.11	-2.17	-2.62
	Others	SYNW1072	methyltransferase	0.64	-1.21	-1.66	-1.53	-1.84	-1.88	-2.33
		SYNW1162 #	hypothetical Arylsulfatase regulator	0.89	-0.97	-1.41	-1.69	-1.86	-2.08	-2.52
		SYNW1214	putative peptidase	0.50	-1.47	-1.76	-1.98	-2.03	-2.05	-2.72
	Hypotheticals	SYNW0055	conserved hypothetical protein	0.98	-1.34	-1.61	-1.83	-1.83	-1.98	-2.49
		SYNW0381 #	hypothetical protein	1.38	-0.58	-1.26	-1.02	-0.68	-0.91	-1.74
		SYNW1139 #	hypothetical protein with DUF4278 [PF14105] domain	1.48	-1.21	-1.30	-1.91	-1.47	-1.52	-2.62
		SYNW1352 ^c #	hypothetical protein with DUF4278 [PF14105] domain	1.94	-1.14	-1.41	-1.53	-1.24	-1.40	-2.22
		SYNW1950 #	hypothetical protein with DUF4278 [PF14105] domain	2.76	-1.38	-1.43	-1.69	-1.78	-1.93	-2.91
		SYNW2450	hypothetical protein with DUF4278 [PF14105] domain	1.25	-0.75	-1.37	-1.47	-1.59	-1.11	-2.01
WH7803	Cell envelope	SynWH7803_2236 ^a	porin,	1.42	-1.56	-2.23	-2.55	-2.79	-2.71	-3.11
	Carbon fixation	SynWH7803_0684 ^b	<i>csoS1/ccmK2</i> : possible CO ₂ concentrating mechanism protein	0.66	-1.08	-1.56	-1.86	-1.47	-2.00	-2.10
	Respiration *	SynWH7803_2016	<i>atpC</i> : ATP synthase gamma chain	1.09	-1.33	-2.13	-2.45	-2.47	-2.50	-2.55
		SynWH7803_2017	<i>atpA</i> : ATP synthase alpha chain	0.72	-1.33	-2.16	-2.48	-2.51	-2.42	-2.30
		SynWH7803_2018	<i>atpD</i> : ATP synthase delta chain	0.76	-1.49	-2.27	-2.62	-2.47	-2.56	-2.60
	Chaperones	SynWH7803_2514	<i>dnaK</i> : Molecular chaperone DnaK	0.98	-1.37	-1.81	-2.37	-2.34	-2.39	-2.44
	Nitrogen transport and assimilation *	SynWH7803_0297 ^d #	<i>amt1</i> : Ammonium transporter	1.16	-0.92	-1.95	-1.96	-2.12	-2.24	-2.26
		SynWH7803_2481 ^e	<i>narB</i> : Nitrate reductase	2.34	0.11	-0.64	-1.40	-0.55	-1.05	-1.17
		SynWH7803_2492	<i>nirA</i> : Ferredoxin--nitrite reductase	2.50	-1.13	-2.33	-2.12	-2.51	-2.57	-2.81
WH8109	Carbon fixation	Syncc8109_1948 ^b	<i>csoS1/ccmK2</i> : possible CO ₂ concentrating mechanism protein	1.18	-0.76	-1.77	-2.13	-2.05	-2.19	-2.47
	Respiration	Syncc8109_1946	<i>ndhD2</i> : proton-translocating NADH-quinone oxidoreductase, chain M	1.22	-0.68	-1.28	-1.76	-1.91	-1.99	-2.45
		Syncc8109_1947	<i>ndhF3</i> : NAD(P)H dehydrogenase, subunit NdhF3 family	1.14	-0.62	-1.29	-1.65	-1.76	-1.79	-2.79
	Hypotheticals	Syncc8109_1164	hypothetical protein	1.21	-0.18	-0.58	-0.51	-0.50	-0.74	-0.50

^{a-k} Homologous genes are indicated by the same letter across tables

*Functional group significantly enriched

#Genes found in genome islands as defined in Dufresne et al. (2008) for WH7803 and WH8102 and in this study for WH8109

§Upregulated in *Prochlorococcus* MED4 infected by P-SSP7 reported in Lindell et al. (2007)

Supplementary Table S10: Host genes with an immediate decline in transcript levels. Log₂ fold change (FC) of infected versus control cultures with time (h) after infection. Positive and negative values indicate an increase and decrease in transcript levels during infection with significant differences in Log₂ FC (q<0.05) shown in red and blue, respectively.

Strain/Host	Functional group	Gene	Predicted function	Log ₂ FC (infected/control)						
				0h	0.5h	1h	1.5h	2h	3h	4h
WH8102	Nitrogen transport and assimilation *	SYNW0253 ^d	<i>amt1</i> : Ammonium transporter family	-0.80	-0.53	-1.68	-1.70	-1.92	-1.98	-1.98
		SYNW0275	<i>ntcA</i> : Global nitrogen transcriptional regulator, CRP family	-3.04	-0.53	-1.46	-1.39	-1.19	-1.55	-0.92
		SYNW0374 #	<i>urtA2</i> : putative urea binding protein	-1.10	-0.32	-1.56	-1.57	-1.66	-1.83	-1.77
		SYNW1073	<i>glnA</i> : Glutamine synthetase,	-0.97	-0.64	-1.83	-1.75	-1.82	-2.01	-2.07
		SYNW1412 #	putative agmatine ureohydrolase	-1.22	-0.06	-0.77	-0.62	-0.65	-0.57	-0.50
		SYNW1414 #	<i>hypB</i> : hydrogenase expression/formation protein HypB	-1.70	0.10	-0.54	-0.56	-0.34	-0.51	-0.50
		SYNW1415 #	ABC transporter, nitrate-like substrate binding protein	-1.99	0.07	-0.67	-0.85	-0.49	-1.07	-1.35
		SYNW1416 #	ABC transporter, nitrate-like	-1.69	0.03	-0.75	-0.59	-0.63	-0.85	-0.56
		SYNW1417 #	ABC transporter, nitrate-like, ATP binding component	-0.92	-0.20	-0.47	-0.72	-0.50	-0.69	-0.80
		SYNW1424 #	conserved hypothetical protein	-0.72	-0.21	-0.92	-0.78	-1.11	-0.91	-1.02
		SYNW1425 #	probable nitrilase	-0.65	-0.21	-0.60	-0.52	-0.64	-0.50	-0.29
		SYNW1426 #	conserved hypothetical protein	-1.16	-0.19	-1.02	-0.94	-0.59	-0.76	-1.43
		SYNW1427 #	predicted flavoprotein involved in K ⁺ transport (COG2072)	-1.65	-0.18	-0.74	-0.90	-0.63	-0.57	-0.57
		SYNW2438	<i>urtE</i> : Putative ATP-binding subunit of urea ABC transport system	-1.38	-0.45	-1.27	-1.23	-1.38	-1.53	-1.29
		SYNW2439	<i>urtD</i> : Putative ATP-binding subunit of urea ABC transport system	-1.35	-0.39	-1.07	-1.44	-1.94	-1.56	-2.14
		SYNW2440	<i>urtC</i> : Putative membrane protein of ABC transport system	-1.39	0.04	-1.08	-1.46	-1.16	-1.24	-1.75
		SYNW2442	<i>urtA1</i> : putative urea ABC transporter, urea binding protein	-1.16	-0.38	-1.73	-1.70	-1.76	-1.90	-1.80
		SYNW2445	<i>ureE</i> : urease accessory protein E	-1.57	-0.36	-0.88	-0.83	-0.92	-0.91	-1.17
		SYNW2446	<i>ureD</i> : urease accessory protein D	-1.06	-0.29	-0.85	-0.70	-0.80	-0.75	-0.67
		SYNW2447	<i>ureA</i> : urease gamma subunit	-1.84	-0.24	-1.30	-1.26	-0.72	-1.42	-0.98
		SYNW2448	<i>ureB</i> : urease beta subunit	-1.83	-0.03	-1.43	-1.10	-1.22	-0.63	-0.79
		SYNW2449	<i>ureC</i> : urease alpha subunit	-1.12	-0.14	-0.96	-0.83	-0.81	-1.11	-1.12
		SYNW2453	<i>asnB</i> : putative asparagine synthetase	-1.41	0.43	-0.61	-0.61	-0.24	-0.88	-0.94
		SYNW2454	<i>pyrB</i> : putative aspartate carbamoyltransferase	-1.49	0.02	-0.75	-0.80	-0.69	-1.10	-1.08
		SYNW2455	sodium:solute transporter family, possible urea transporter	-1.36	-0.20	-1.08	-1.03	-0.82	-1.15	-1.19
		SYNW2456	conserved hypothetical protein	-2.36	-0.17	-0.92	-0.78	-0.89	-0.83	-1.07
		SYNW2457	conserved hypothetical protein	-1.13	0.13	-0.83	-0.63	-0.46	-1.18	-1.26
		SYNW2458	possible transglutaminase-like enzyme	-1.00	-0.08	-0.93	-1.04	-0.68	-0.98	-1.07
		SYNW2462	nitrate transporter, MFS family	-1.16	-0.35	-1.68	-1.68	-1.82	-1.98	-1.73
		SYNW2463	nitrate transporter, MFS family	-1.07	-0.27	-1.64	-1.65	-1.73	-1.88	-1.70
		SYNW2464 ^e	<i>narB</i> : nitrate reductase	-1.72	0.32	-0.69	-0.83	-0.59	-0.53	-0.53
		SYNW2490 #	<i>cynS</i> : cyanate lyase	-1.94	-0.09	-0.73	-0.79	-0.54	-1.16	-1.38
		SYNW2492 #	glutamine amidotransferase-phosphoribosyl anthranilate transferase	-2.08	-0.32	-1.24	-0.93	-0.90	-1.32	-1.43
	Translation	SYNW1962	<i>eIF2B</i> : putative translation initiation factor eIF-2B alpha subunit	-0.65	-0.53	-0.83	-1.14	-1.21	-1.05	-0.95
		SYNW0562	leucine aminopeptidase	-0.82	-0.78	-1.48	-1.29	-1.54	-1.36	-1.29
	Hypotheticals	SYNW0802	conserved hypothetical	-2.92	-0.37	-0.78	-1.06	-0.63	-1.62	-1.17
		SYNW1141 #	hypothetical protein	-2.75	-0.32	-1.24	-1.06	-0.79	-1.60	-0.97
		SYNW1357 #	conserved hypothetical protein	-3.72	-0.74	-1.77	-1.91	-1.71	-2.84	-1.20
		SYNW2404 #	conserved hypothetical protein	-1.26	-0.13	-1.16	-0.88	-0.61	-0.93	-1.12
		SYNW2436	conserved hypothetical protein	-2.25	-0.26	-0.84	-1.12	-1.03	-1.33	-1.53
		SYNW2437 ^f	conserved hypothetical protein	-1.61	0.02	-0.50	-0.97	-0.09	-1.16	-1.00
WH7803	Energy metabolism	SynWH7803_0029	<i>gap1</i> : Glyceraldehyde-3-phosphate dehydrogenase, glycolysis	-0.94	-1.52	-2.06	-2.40	-2.42	-2.46	-2.48
	Transport and binding proteins	SynWH7803_2499	permease,of the major facilitator superfamily	-0.58	-0.70	-1.07	-1.41	-1.00	-1.21	-1.08
WH8109		Syncc8109_1616 #	conserved hypothetical protein	-1.24	-0.40	-0.77	-0.89	-0.90	-1.11	-1.42
		Syncc8109_2674 ^f	conserved hypothetical protein	-2.87	-0.57	-1.28	-2.09	-0.65	-2.54	-0.30
		Syncc8109_2709	conserved hypothetical protein	-3.48	-0.90	-1.36	-2.25	-1.06	-2.34	-0.40

^{a-k} Homologous genes are indicated by the same letter across tables

*Functional group significantly enriched

#Genes found in genome islands as defined in Dufresne et al. (2008) for WH7803 and WH8102 and in this study for WH8109

Supplementary Table S11: Host genes with a subsequent increase in transcript levels. Log₂ fold change (FC) of infected versus control cultures with time (h) after infection. Positive and negative values indicate an increase and decrease in transcript levels during infection with significant differences in Log₂ FC (q<0.05) shown in red and blue, respectively.

				Log ₂ FC (infected/control)						
Strain/Host	Functional group	Gene	Predicted function	0h	0.5h	1h	1.5h	2h	3h	4h
WH8102	Cell envelope	SYNW2107 #	conserved hypothetical protein with SecA_DEAD [PF07517] domain	0.01	0.82	0.47	0.20	1.27	-0.05	0.09
	DNA repair	SYNW1575 #	hypothetical protein with AAA_25 [PF13481] domain	0.00	0.25	0.70	0.18	-0.27	-0.33	-0.15
	Other	SYNW1659 #	conserved hypothetical protein with DUF3378 [PF1867] domain [this domain is found at the C-terminus of type I restriction enzymes]	0.15	0.12	0.25	0.71	-0.46	-0.70	-0.76
	Hypotheticals	SYNW0164	hypothetical protein	-0.08	0.78	0.39	0.41	0.32	0.10	-0.27
		SYNW1101 #	hypothetical protein	-0.19	0.50	0.22	0.28	0.17	0.16	0.06
		SYNW1573 #	hypothetical protein	0.15	0.55	0.26	0.50	0.28	0.30	0.00
		SYNW1574 #	conserved hypothetical protein	0.29	0.27	0.50	0.00	0.45	-0.05	0.11
		SYNW1658 ^g #	conserved hypothetical protein with DUF1651 [PF07864] domain	-0.07	0.14	0.59	0.95	0.61	0.59	0.78
		SYNW2102 #	hypothetical protein	-0.23	0.32	0.64	0.61	0.69	-0.21	-0.30
		SYNW2106 ^g # §	conserved hypothetical protein with DUF1651 [PF07864] domain	0.17	0.41	0.39	0.33	0.72	0.68	0.09
		SYNW2413 #	hypothetical protein	0.12	0.51	0.37	-0.13	-0.14	0.37	-0.02
		SYNW2415 #	hypothetical protein	0.04	-0.07	0.06	0.01	0.69	-0.19	-0.42
WH7803	DNA repair	SynWH7803_1081 ^h §	<i>umuD</i> : Bacterial UmuD protein homolog (SOS-regulated protein)	-0.14	0.47	0.52	0.50	0.27	0.04	0.04
	Other	SynWH7803_0111	atypical alkaline phosphatase with phytase-like insertion	-0.02	0.55	-0.06	-0.30	-0.36	-0.17	-0.41
		SynWH7803_1790 #	uncharacterized conserved secreted protein, pili subunit superfamily	0.26	0.06	-0.04	0.57	0.29	0.39	0.35
	Hypotheticals	SynWH7803_0846 #	conserved hypothetical protein	0.15	0.16	0.01	-0.39	-0.39	-0.80	0.57
		SynWH7803_0900 #	conserved hypothetical protein	0.07	1.05	0.04	-0.32	-0.45	-0.43	-1.21
WH8109	Hypotheticals	Syncc8109_0530	conserved hypothetical protein	0.19	-0.46	0.15	1.02	0.32	0.17	0.68
		Syncc8109_0533	hypothetical protein	0.09	-0.02	1.54	1.87	1.86	1.96	1.44
		Syncc8109_0594 #	hypothetical protein	-0.17	-0.07	-0.28	0.17	0.71	0.45	0.64
		Syncc8109_1070	hypothetical protein	-0.03	-0.15	0.46	0.20	0.76	0.04	0.06
		Syncc8109_1509	conserved hypothetical protein	0.12	-0.51	-0.89	0.02	-0.04	-0.26	0.75
		Syncc8109_2306 #	hypothetical protein	-0.84	0.09	1.83	0.54	0.31	-0.38	-1.91
		Syncc8109_2418	hypothetical protein	0.27	-0.13	1.35	1.81	1.59	1.01	0.77
		Syncc8109_2422	conserved hypothetical protein	0.13	-0.46	-0.04	1.03	0.32	0.33	0.57

^{a-k} Homologous genes are indicated by the same letter across tables
#Genes found in genome islands as defined in Dufresne et al. (2008) for WH7803 and WH8102 and in this study for WH8109
§Upregulated in *Prochlorococcus* MED4 infected by P-SSP7 reported in Lindell et al. (2007)

Supplementary Table S12: Host genes with unchanged transcript levels for 2-4 h after infection. Log₂ fold change (FC) of infected versus control cultures with time (h) after infection. Positive and negative values indicate an increase and decrease in transcript levels during infection with significant differences in Log₂ FC (q<0.05) shown in red and blue, respectively.

				Log ₂ FC (infected/control)						
Strain/Host	Functional group	Gene	Predicted function	0h	0.5h	1h	1.5h	2h	3h	4h
WH8102	Cell envelope	SYNW0943	<i>rlpA</i> : Bacterial septal ring protein RlpA homolog	-0.19	-0.42	-0.49	-0.64	-0.38	-0.64	-0.41
	DNA repair	SYNW0366 #	hypothetical protein with AAA_25 [PF13481] domain	0.24	-0.01	-0.20	-0.09	-0.03	-0.03	-0.22
		SYNW1044 ^h	<i>umuD</i> : Bacterial UmuD protein homolog (SOS-regulated protein)	-0.36	-0.36	-0.31	-0.17	-0.11	-0.42	-0.02
		SYNW2412 #	conserved hypothetical protein with AAA_25 [PF13481] domain	0.30	0.35	0.13	-0.03	0.22	0.06	0.19
	Transport and binding proteins	SYNW0147	putative cation (Mg/Co) transporter, MIT family	-0.25	-0.26	-0.06	-0.04	0.03	-0.55	-0.47
		SYNW0427 #	possible multidrug efflux ABC transporter	0.31	0.12	-0.23	-0.31	-0.39	-1.27	-1.95
		SYNW1685	putative SMR family multidrug efflux transporter	-0.05	-0.42	-0.44	-0.23	-0.22	-0.51	-0.55
		SYNW1701	possible mechanosensitive ion channel, MscS family	-0.04	-0.23	-0.11	0.03	-0.27	-0.41	-0.21
		SYNW2485 #	putative cyanate ABC transporter	-0.56	0.43	-0.28	-0.88	-0.32	-1.54	-2.14
		SYNW2486 #	putative cyanate ABC transporter	-0.50	0.28	-0.57	-0.77	-0.32	-1.32	-1.20
	Molecular chaperones	SYNW2365 ⁱ	putative small heat shock protein	0.26	-0.18	-0.34	-0.16	0.23	0.03	-0.10
	Metabolism of cofactors and vitamins	SYNW2306	<i>menC</i> : putative O-succinylbenzoate synthase	-0.36	-0.34	-0.23	-0.25	-0.31	-0.43	-0.24
	Respiration	SYNW1860 ^j	<i>ctaA</i> : cytochrome C oxidase assembly protein	0.11	-0.03	0.00	-0.09	-0.22	-0.26	-0.88
		SYNW1861 ^k	<i>ctaC</i> , <i>coxB</i> : possible cytochrome c oxidase subunit II	0.27	-0.16	0.05	-0.28	0.01	-0.65	-1.96
	Amino acid biosynthesis	SYNW2452	putative N-carbamoyl-L-amino-acid hydrolase	-1.73	-0.01	-0.82	-0.79	-0.56	-0.83	-0.94
	Other	SYNW0071	conserved hypothetical protein with PIN [PF01850] domain	0.36	-0.21	-0.37	-0.19	-0.31	-0.47	-0.61
		SYNW0312	conserved hypothetical protein with two phospholipase PLDc_2 [PF13091] domains	-0.02	0.15	-0.25	-0.23	-0.05	-0.21	-0.18
		SYNW0375 #	conserved hypothetical protein with PAP_fibrillin [PF04755] region	-0.13	-0.25	-0.16	-0.44	0.09	-0.82	-1.08
		SYNW0539	similar to stage II sporulation protein D	-0.10	-0.34	-0.30	-0.08	-0.17	-0.21	-0.40
		SYNW0978	16S ribosomal RNA methyltransferase RsmE	-0.17	0.04	0.05	0.15	0.60	0.07	-0.16
		SYNW1569 #	possible phage integrase	0.17	-0.08	-0.25	-0.03	-0.27	-0.41	-0.37
		SYNW1578 #	Staphylococcal nuclease homologue	0.04	-0.07	0.13	-0.31	0.10	-0.40	-1.00
		SYNW1665 #	putative phage transcriptional regulator AlpA	-0.27	0.25	0.08	0.15	0.07	-0.86	-1.14
		SYNW1946 #	PhoH family protein with PIN_4 (PF13638) domain	-0.82	0.53	-0.17	-0.07	0.36	-0.66	-0.08
		SYNW2103 #	putative AbrB- like transcriptional regulator	0.61	-0.01	-0.03	0.01	0.78	-0.52	-1.47
		SYNW2409 #	putative hemolysin-type calcium-binding protein; similar to HlyA	0.11	-0.53	-0.27	-0.37	-0.48	-1.49	-2.30
		SYNW2488 #	MttA_Hcf106 superfamily translocase protein	0.24	0.11	0.15	-0.23	-0.23	-1.87	-3.33
	Hypotheticals	SYNW0042	conserved hypothetical protein	0.01	-0.42	-0.70	-0.65	-0.53	-1.11	-1.12
		SYNW0347 #	conserved hypothetical protein	0.28	0.31	-0.35	-0.20	-0.56	-1.36	-1.60
		SYNW0373 #	hypothetical protein	-0.04	-0.23	0.29	-0.24	0.42	-0.68	-0.34
		SYNW0377 #	hypothetical protein	-0.13	-0.39	0.19	-0.27	-0.33	-0.49	-0.48
		SYNW0429 #	hypothetical protein	-0.08	0.57	0.40	-0.11	-0.13	-1.11	-1.57
		SYNW0433 #	hypothetical protein	0.26	-0.18	0.05	-0.40	-0.51	-1.35	-2.02
		SYNW0922	hypothetical protein	0.15	-0.63	-0.83	-0.66	-0.71	-1.35	-1.70
		SYNW0961 #	hypothetical protein	-0.10	0.19	0.43	-0.27	0.18	-0.59	-0.90
		SYNW0972	conserved hypothetical protein	0.04	-0.32	-0.34	-0.26	-0.10	-0.27	-0.41
		SYNW1061	hypothetical protein	0.37	0.13	-0.49	-0.47	-0.20	-1.51	-1.81
		SYNW1102 #	conserved hypothetical protein	0.13	0.04	0.17	-0.07	-0.21	-0.14	-0.28
		SYNW1140 # §	conserved hypothetical protein	0.04	-0.38	-0.26	0.05	-0.38	-0.65	-0.41
		SYNW1142 #	hypothetical	-0.17	0.06	-0.40	-0.39	0.54	0.00	0.13
		SYNW1163 #	conserved hypothetical protein, phage associated	-0.05	0.23	0.42	0.15	0.56	-0.03	-0.39
		SYNW1184 #	hypothetical protein	0.28	0.31	0.16	-0.17	-0.79	-1.12	-1.63
		SYNW1350 #	hypothetical protein	0.11	-0.08	0.20	-0.10	0.07	-0.76	-0.91
		SYNW1374 #	hypothetical protein	-0.06	-0.29	-0.51	-0.44	-0.42	-0.45	-1.21
		SYNW1552 #	hypothetical protein	0.08	-0.36	-0.37	-0.37	-0.22	-0.47	-0.74
		SYNW1562 #	conserved hypothetical protein with DUF3303 [PF11746] domain	0.03	0.10	-0.26	-0.08	-0.04	-0.86	-2.78
		SYNW1563 #	conserved hypothetical protein	0.17	0.14	0.16	-0.21	-0.24	-0.96	-1.67
		SYNW1576 #	conserved hypothetical protein	-0.14	-0.26	-0.32	-0.08	-0.02	-0.38	-0.47
		SYNW1656 #	hypothetical protein	-0.39	0.32	0.16	-0.34	-0.32	-0.99	-0.93

Strain/Host	Functional group	Gene	Predicted function	Log ₂ FC (infected/control)						
				0h	0.5h	1h	1.5h	2h	3h	4h
		SYNW1672	hypothetical protein	0.09	0.31	-0.40	-0.26	-0.24	-0.56	-0.02
		SYNW1829	conserved hypothetical protein with DUF1643 [PF07799]	-0.05	0.08	-0.23	-0.34	0.05	-0.12	-0.19
		SYNW1944 ^g #	conserved hypothetical protein with DUF1651 [PF07864] domain	0.13	-0.30	-0.25	-0.07	-0.03	-0.93	-0.98
		SYNW1948 #	hypothetical protein	0.07	0.08	-0.06	0.27	-0.39	-0.38	-0.84
		SYNW2101 #	hypothetical protein	0.15	0.44	0.06	0.13	0.13	0.06	-0.30
		SYNW2112 #	hypothetical protein	0.01	-0.20	-0.26	-0.23	-0.33	-0.82	-0.97
		SYNW2451	conserved hypothetical protein with DUF1028 [PF06267] domain	-0.05	0.12	-0.33	-0.54	-0.31	-1.22	-1.03
		SYNW2514	conserved hypothetical protein	-0.33	-0.34	-0.12	0.03	0.42	-0.74	-0.19
WH7803	Cell envelope	SynWH7803_0595	conserved hypothetical protein with transmembrane domains	0.04	-0.23	0.12	-0.18	-0.08	-0.24	-0.35
		SynWH7803_0908 #	uncharacterized membrane protein	0.23	0.56	0.13	-0.54	0.49	0.14	-0.27
		SynWH7803_0918 #	uncharacterized membrane protein	-0.10	-0.24	-0.14	-0.37	-0.05	-0.31	-0.04
		SynWH7803_0991 #	hypothetical protein with transmembranes domains	0.32	0.31	0.29	-0.04	-0.31	-0.66	-0.28
		SynWH7803_1649	uncharacterized membrane protein	0.23	-0.17	-0.40	-0.35	-0.65	0.25	-0.07
		SynWH7803_1661	uncharacterized conserved membrane protein	-0.10	0.20	0.20	0.39	0.09	-0.04	0.08
		SynWH7803_1672 #	uncharacterized membrane protein	0.70	0.49	-0.23	-0.32	-0.82	-0.52	-0.80
		SynWH7803_1674 #	uncharacterized membrane protein	0.09	0.33	0.56	0.13	-0.47	0.08	-0.41
		SynWH7803_1796 #	conserved secreted protein, pili subunit superfamily	-0.18	0.41	0.52	-0.32	-0.33	-1.56	-1.39
		SynWH7803_1797 #	conserved secreted protein	-0.13	0.28	0.12	-0.64	-0.62	-1.37	-1.48
		SynWH7803_1798 #	conserved secreted protein, pili subunit superfamily	0.30	0.33	0.26	-0.30	-0.74	-1.20	-1.30
		SynWH7803_1799 #	conserved secreted protein, pili subunit superfamily	0.12	0.21	-0.03	-0.70	-0.55	-1.23	-1.30
		SynWH7803_1800 #	uncharacterized conserved membrane protein	0.14	0.08	0.14	-0.19	-0.54	-1.26	-1.21
		SynWH7803_2116 #	uncharacterized membrane protein	0.58	0.35	0.10	-0.40	-0.58	-0.68	-0.64
		SynWH7803_2448	conserved hypothetical membrane protein (putative metal-binding protein)	0.10	-0.14	0.06	-0.24	-0.41	-0.06	-0.18
	DNA repair	SynWH7803_0163	<i>mutY</i> : A/G-specific DNA glycosylase MutY	0.11	-0.26	-0.30	-0.27	-0.15	-0.55	-0.26
		SynWH7803_1080	<i>umuC</i> : Bacterial UmuC protein homolog (SOS-regulated protein)	0.02	0.28	0.14	-0.01	0.19	-0.08	-0.05
		SynWH7803_1507	<i>recC</i> : Exodeoxyribonuclease V gamma chain	-0.03	-0.02	-0.26	-0.23	-0.42	-0.59	-0.03
	Molecular chaperones	SynWH7803_1632	DNA or RNA helicases of superfamily II	-0.11	-0.11	-0.22	-0.23	-0.23	-0.25	-0.68
		SynWH7803_0904 #	DnaJ domain containing protein	0.05	-0.47	-0.31	-0.52	-0.60	-0.62	-0.61
		SynWH7803_2403 ⁱ	putative small heat shock protein	0.10	0.09	-0.02	-0.18	-0.38	-0.60	-0.61
	Metabolism of cofactors and vitamins	SynWH7803_0083	<i>cobH</i> : Precorrin-8X methylmutase	-0.05	-0.33	-0.12	-0.05	-0.01	-0.29	0.01
	Transport and binding proteins	SynWH7803_0989 #	<i>znuC</i> : ABC-type Mn2+/Zn2+ transport system permease component	0.21	0.19	-0.29	-0.12	-0.03	-0.08	-0.05
		SynWH7803_1228	<i>sphX</i> : ABC transporter, substrate binding protein, phosphate	0.58	0.43	0.05	-0.51	-0.66	-0.96	-1.28
	Carbohydrate metabolism	SynWH7803_0166	<i>xylB</i> : Xylulose kinase	0.01	0.05	-0.09	0.04	0.10	-0.15	-0.01
		SynWH7803_1737 #	predicted sugar kinase fused to an uncharacterized domain	-0.14	-0.08	-0.20	-0.17	-0.30	-0.27	0.01
		SynWH7803_2527	glycosyltransferase of family GT4; possible sucrose-phosphate synthase	0.03	-0.19	-0.38	-0.76	-0.58	-0.97	-0.90
	Amino acids metabolism	SynWH7803_1326	<i>proA</i> : Gamma-glutamyl phosphate reductase	0.31	-0.34	-0.85	-0.87	-0.61	-0.67	0.30
		SynWH7803_2471	probable aminotransferase	-0.04	-0.17	-0.37	-0.51	-0.44	-0.64	-0.53
	Respiration	SynWH7803_1869	<i>ctaB</i> : Protoheme IX farnesyltransferase	-0.32	0.20	-0.10	-0.54	-0.26	-0.29	-0.35
		SynWH7803_1870 ^j	<i>ctaA</i> : cytochrome C oxidase assembly protein	0.00	0.06	0.07	-0.35	0.05	-0.22	-0.24
		SynWH7803_1871 ^k	<i>ctaC</i> , <i>coxB</i> : Cytochrome c oxidase subunit II	0.06	0.49	0.12	-0.33	-0.60	-0.78	-1.51
	Secreted proteins	SynWH7803_1872	<i>ctaD</i> , <i>coxA</i> : Cytochrome c oxidase subunit I	0.18	0.03	-0.27	-0.36	-0.44	-0.61	-0.72
		SynWH7803_1478	uncharacterized conserved secreted protein	-0.08	0.10	0.12	-0.25	-0.72	-1.19	-1.35
		SynWH7803_2111 #	uncharacterized secreted protein	0.54	-0.22	-0.25	-0.55	-0.52	-1.23	-1.24
		SynWH7803_2358	uncharacterized conserved secreted protein	0.00	0.45	0.48	0.37	0.20	-0.18	-0.20
	Others	SynWH7803_0090 #	possible glycosyltransferase	0.51	0.28	0.70	-0.23	-0.28	-1.20	-1.05
		SynWH7803_0144	dienelactone hydrolase family protein	0.07	-0.14	-0.29	-0.10	-0.23	-0.20	-0.57
		SynWH7803_0447	possible metal-dependent protease of the PAD1/JAB1 superfamily	0.35	-0.47	-0.44	-0.40	-0.15	0.10	0.26
		SynWH7803_0464	<i>pex</i> : Possible Pex protein (Period-extender gene)	-0.26	-0.04	0.20	-0.37	-0.01	-0.64	-0.48
		SynWH7803_0809	predicted NAD/FAD-dependent oxidoreductase	-0.01	-0.31	-0.38	-0.30	-0.02	-0.27	-0.10
		SynWH7803_0876 #	predicted esterase	-0.01	-0.07	-0.23	-0.29	-0.23	-0.30	-0.70

Strain/Host	Functional group	Gene	Predicted function	Log ₂ FC (infected/control)						
				0h	0.5h	1h	1.5h	2h	3h	4h
		SynWH7803_0885	possible thiopurine S-methyltransferase	-0.09	-0.29	-0.28	-0.24	-0.07	-0.03	-0.24
		SynWH7803_1002 #	possible dehydrogenase	0.00	-0.13	-0.27	0.08	-0.21	-0.06	-0.06
		SynWH7803_1441§	<i>hli</i> : High light inducible protein	0.10	0.06	0.00	-0.43	-0.85	-0.97	-1.20
		SynWH7803_1442	<i>ubiH</i> : Putative 2-octaprenyl-6-methoxyphenol hydroxylase	0.12	0.10	-0.03	-0.43	-0.25	-0.80	-0.75
		SynWH7803_2417	<i>desC</i> : Fatty-acid desaturase	-0.02	0.01	-0.06	-0.37	-0.38	-0.60	-0.65
	Hypotheticals	SynWH7803_0079	conserved hypothetical protein	-0.30	-0.19	-0.19	-0.32	-0.22	0.33	-0.03
		SynWH7803_0158	hypothetical protein	-0.37	0.07	-0.45	-0.75	-0.47	0.32	-0.77
		SynWH7803_0215	conserved hypothetical protein with DUF1537 [PF07005] domain	0.01	-0.28	-0.32	-0.23	-0.17	0.06	-0.15
		SynWH7803_0271	conserved hypothetical protein with SIMPL(DUF541) [PF04402] domain	-0.07	-0.20	-0.25	-0.33	-0.34	-0.40	-0.39
		SynWH7803_0394	hypothetical protein	-0.15	0.19	0.44	0.37	0.26	-0.45	0.11
		SynWH7803_0724	conserved hypothetical protein	-0.08	-0.38	-0.12	-0.52	-0.37	-0.15	-0.50
		SynWH7803_0818	conserved hypothetical protein	-0.26	0.09	-0.52	-0.43	-0.62	-0.80	-0.24
		SynWH7803_0883 #	hypothetical protein	0.23	0.05	0.17	-0.47	-0.44	-0.40	-0.61
		SynWH7803_0897 #	hypothetical protein	0.07	0.39	-0.09	-0.76	-0.70	-1.15	-1.04
		SynWH7803_0902 #	hypothetical protein	0.93	-0.79	-0.78	-0.33	-0.42	-0.84	-1.32
		SynWH7803_0903 #	hypothetical protein	0.00	0.48	-0.14	-0.11	-0.34	-0.46	-0.81
		SynWH7803_0907 #	hypothetical protein	0.07	-0.06	-0.85	-1.18	-0.59	-0.66	-0.92
		SynWH7803_1012 #	hypothetical protein	0.02	-0.33	-0.46	-0.13	-0.32	-0.39	-0.80
		SynWH7803_1044	conserved hypothetical protein	-0.02	0.36	-0.38	-0.22	-0.54	-0.12	-1.08
		SynWH7803_1322	conserved hypothetical protein	0.20	-0.32	-0.69	-0.62	-0.25	-0.42	-0.47
		SynWH7803_2086	conserved hypothetical protein	-0.10	-0.14	-0.15	-0.52	-0.43	-0.75	-1.03
		SynWH7803_2117 #	hypothetical protein	0.51	-0.02	-0.01	-0.89	-0.26	-0.54	-0.49
		SynWH7803_2367	conserved hypothetical protein	0.10	-0.14	-0.21	-0.06	-0.24	-0.18	-0.61
WH8109	Cell envelope	Syncc8109_1896	competence-related protein pseudogene	-0.04	-0.19	-0.21	-0.02	-0.10	-0.51	0.60
	DNA repair	Syncc8109_0511	<i>smf</i> : putative DNA processing protein (Smf family) pseudogene	0.40	-0.28	-0.26	-0.47	-0.42	-0.30	-0.08
	Molecular chaperones	Syncc8109_2605 ⁱ	putative small heat shock protein	-0.39	-0.03	-0.28	0.15	0.15	0.11	-0.13
	Transcription	Syncc8109_0081	<i>mesJ</i> : possible MesJ protein	-0.11	-0.19	0.06	-0.25	-0.26	-0.68	0.21
	Hypotheticals	Syncc8109_0402 #	hypothetical protein	-0.42	-0.26	-0.41	-0.74	-1.00	-1.40	-1.93
		Syncc8109_0449 #	hypothetical protein	-0.22	-0.88	0.51	-0.19	-0.56	-0.96	-1.78
		Syncc8109_0454 #	hypothetical protein	-0.49	-0.72	-0.72	-0.41	-1.06	-1.32	-2.10
		Syncc8109_0491 ^g	conserved hypothetical protein with DUF1651 [PF07864] domain	-0.19	-0.21	-0.25	-0.36	-0.19	-0.91	0.19
		Syncc8109_0492	hypothetical protein	0.12	-0.42	-0.37	-0.18	-0.46	-0.05	-0.77
		Syncc8109_0577 #	hypothetical protein	0.05	-0.21	0.07	-0.01	-0.08	0.25	-0.69
		Syncc8109_0582 #	conserved hypothetical protein	2.14	-0.86	-0.89	-1.10	-1.03	-1.47	-2.04
		Syncc8109_0597 #	conserved hypothetical protein	-0.12	0.11	-0.30	-0.30	-0.16	-0.54	-0.39
		Syncc8109_0729	hypothetical protein	-0.24	-0.19	0.00	0.35	0.16	0.01	-0.11
		Syncc8109_0797	hypothetical protein	-0.20	-0.33	-0.24	-0.54	-0.50	-0.29	-0.68
		Syncc8109_0906 #	conserved hypothetical protein	-0.16	-0.26	-0.31	-0.02	-0.11	-0.29	-0.02
		Syncc8109_0938 #	hypothetical protein	-0.19	-0.10	-0.24	-0.23	-0.38	-0.47	-0.73
		Syncc8109_0990	conserved hypothetical protein	0.01	-0.05	-0.07	-0.27	-0.36	-0.38	-0.07
		Syncc8109_1046 #	hypothetical protein	0.20	-0.20	0.33	-0.10	-0.39	-0.64	-0.03
		Syncc8109_1052 #	conserved hypothetical protein	-0.47	-0.19	-0.05	-0.33	-0.67	-1.53	-1.04
		Syncc8109_1083	hypothetical protein	-0.05	-0.50	-0.35	-0.54	-0.32	-0.39	-0.36
		Syncc8109_1088	hypothetical protein	-0.23	-0.20	-0.32	-0.27	0.22	-0.77	-0.16
		Syncc8109_1108	conserved hypothetical protein	-0.63	0.16	0.10	0.05	-0.15	-0.59	0.28
		Syncc8109_1112	conserved hypothetical protein	0.15	0.12	-0.54	-0.11	-0.31	0.07	-0.26
		Syncc8109_1118	conserved hypothetical protein	0.19	-0.14	0.32	0.32	0.44	0.01	0.04
		Syncc8109_1166 ^c	conserved hypothetical protein with DUF4278 [PF14105] domain	2.00	-0.20	-0.68	-0.83	-0.61	-0.90	-1.50
		Syncc8109_1366 #	conserved hypothetical protein	-0.75	-0.17	-0.07	-0.49	-0.59	-1.13	-1.44
		Syncc8109_1424	conserved hypothetical protein	0.04	-0.01	-0.14	-0.09	-0.19	-0.41	0.28
		Syncc8109_1687	hypothetical protein	-0.27	0.09	-0.38	0.10	-0.16	-0.46	-0.81
		Syncc8109_1785 #	conserved hypothetical protein	-0.26	-1.07	-0.60	-0.45	-0.75	-1.37	-1.96
		Syncc8109_1799 #	conserved hypothetical protein	-0.08	-0.39	-0.54	-0.62	-0.51	-0.62	-0.68
		Syncc8109_1807 #	hypothetical protein	-0.47	0.12	-0.60	-0.52	-0.61	-0.81	-1.13
		Syncc8109_1921	hypothetical protein	-0.45	-0.23	-0.56	-0.72	-0.69	-1.27	-1.10
		Syncc8109_1928	hypothetical protein	0.04	0.49	-0.24	-0.10	-0.64	-1.02	-2.02
		Syncc8109_2003	conserved hypothetical protein	0.23	0.05	-0.25	-0.15	-0.51	-0.01	0.19
		Syncc8109_2058 #	hypothetical protein	0.04	-0.14	0.68	0.62	0.57	-0.17	-1.19
		Syncc8109_2318 #	hypothetical protein	-0.07	-0.26	-0.21	-0.07	-0.41	-1.45	-1.32
		Syncc8109_2327 #	hypothetical protein	-0.23	0.33	-0.15	-0.53	-0.26	-0.60	-1.33
		Syncc8109_2334 #	hypothetical protein	-0.10	0.00	-0.12	0.53	-0.60	-0.52	0.34

Strain/Host	Functional group	Gene	Predicted function	Log ₂ FC (infected/control)						
				0h	0.5h	1h	1.5h	2h	3h	4h
		Sync8109_2447 #	hypothetical protein	-0.07	-0.64	-0.26	-0.34	-0.23	-0.15	-0.17
		Sync8109_2626	hypothetical protein	-0.21	-0.06	-0.27	-0.43	-0.31	-0.10	-0.32
		Sync8109_2631	conserved hypothetical protein	0.01	-0.10	-0.19	-0.15	-0.18	-0.35	-0.06
		Sync8109_2653	hypothetical protein	-0.30	0.03	0.24	-0.16	0.20	-0.09	-0.36
		Sync8109_2705	hypothetical protein	-0.40	-0.68	-0.41	-0.22	-0.54	-0.57	-0.92
		Sync8109_2729	conserved hypothetical protein	-1.58	-0.26	-0.87	-0.72	-0.38	-0.76	0.07

^{a-k} Homologous genes are indicated by the same letter across tables; #Genes found in genome islands as defined in Dufresne et al. (2008) for WH7803 and WH8102 and in this study for WH8109; §Upregulated in *Prochlorococcus* MED4 infected by P-SSP7 (Lindell *et al.*, 2007)

Supplementary Table S13: Location and composition of genomic islands in *Synechococcus* WH8109

ISL Name	Start in nt	Stop in nt	Length (bp)	First gene ¹	Last gene ¹	No. of genes	Predicted Protein Product ²	Comparison to CC9605 ³
ISL1	75571	82685	7115	0089	0093	5	Possible bacterial sugar transferase, possible transporter component, putative multidrug efflux ABC transporter	insertion in WH8109
ISL2	85475	103997	18522	0095	0107	12	3 putative glycosyltransferase genes, putative acyltransferase, Hemolysin-type calcium-binding region, ABC transporter, ATP binding protein, serine acetyltransferase	insertion in WH8109
ISL3	163381	168230	4850	0176	0178	3	putative glycosyltransferase	different genes to CC9605, not defined as ISL in CC9605 ³
ISL4	356554	358862	2309	0402		1		insertion in WH8109
ISL5	384801	412483	27682	0435	0472	38	Phycobilisome island: Phycobilin lyase beta subunit, Phycobilin lyase alpha subunit, R-phycocyanin II alpha subunit, phycocyanin beta subunit, phycoerythrobilin:ferredoxin oxidoreductase, 15,16 dihydrobiliverdin:ferredoxin oxidoreductase, C-phycoerythrin class I beta chain, C-phycoerythrin class I alpha chain, putative bilin biosynthesis protein (CpeY), putative bilin biosynthesis protein cpeZ, transcriptional regulator, araC family domain protein, PBS lyase HEAT-like repeat, C-phycoerythrin class II gamma chain linker polypeptide, class II phycoerythrin alpha-subunit, C-phycoerythrin class II beta chain, PBS lyase HEAT-like repeat, possible phycobilisome linker polypeptide, phycobilisome linker polypeptide, phycobilisome linker polypeptide, possible phycobilisome rod-core linker polypeptide (L-RC 28.5), possible phycobilisome polypeptide	mostly same genes as PBS ISL in CC9605 ³
ISL6	482357	508314	25957	0573	0623	50	Band 7 protein, Large exoprotein involved in heme utilization or adhesion	mostly same genes as ISL2 in CC9605 ³
ISL7	665565	671308	5744	0842	0848	7	Phage integrase family domain protein	insertion in WH8109
ISL8	711789	747871	36082	0894	0962	70	putative glutathione S-transferase, cell death suppressor protein Lls1-like, Glucosylglycerol 3-phosphatase, possible L-asparaginase II, putative ferric uptake regulator FUR family, possible ICC (3',5'-cyclic-nucleotide phosphodiesterase) protein, putative histone deacetylase/AcuC/AphA family protein, possible high light inducible protein, putative pectate lyase, possible carbonic anhydrase, gid protein, putative cation efflux transporter (CDF family), carotenoid isomerase, and 2 tRNAs	mostly same genes as in ISL3 in CC9605 ³
ISL9	816293	832949	16657	1040	1061	21	chlorophyll a/b binding light harvesting protein PcbD, HTH of cAMP family transcriptional regulator, Flavodoxin long chain, ferritin, ABC-type Fe3+ transport system periplasmic component, 2OG-Fe(II) oxygenase superfamily enzyme, possible porin	mostly same genes as ISL12 in CC9605 ³
ISL10	1029445	1038893	9449	1365	1382	18	ABC transporter, substrate binding protein phosphate, putative transcriptional regulator Crp/Fnr family	insertion in WH8109
ISL11	1132215	1136814	4600	1497	1508	12	phage capsid family protein	insertion in WH8109
ISL12	1186415	1217764	31350	1570	1651	81	Chromate transporter, putative arsenite transporter ACR3 family, zeta-carotene desaturase-like, Heat shock protein DnaJ-like, short-chain dehydrogenase/reductase (SDR) superfamily, possible ATLS1-like light-inducible protein, 30S ribosomal protein S21, membrane protein, possible transcriptional regulator, photosystem II D1 protein form I, cation diffusion facilitator family transporter	mostly same genes as in CC9605, ISL7 in CC9605 ²
ISL13	1274771	1328077	53306	1722	1814	91	3 putative phage-related proteins, Subtilisin-like serine proteases-like, putative phycobilisome linker polypeptide, possible phage integrase family, possible fatty acid desaturase, nuclease (SNase-like), Dps protein family protein, LexA repressor, and tRNA	ISL in CC9605 contains different genes
ISL14	1527273	1531105	3833	2058	2062	3		insertion in WH8109
ISL15	1720529	1745215	24686	2290	2334	41	weak similarity to phage integrase family, putative methyltransferase type 11, Band 7 protein, Band 7 protein, possible integrase/recombinase	ISL in CC9605 contains different genes
ISL16	1774547	1779312	4766	2374	2382	8	Phospholipase D/Transphosphatidylase, phosphoglucomutase	mostly same genes as ISL17 in CC9605 ³
ISL17	1833127	1837140	4014	2447	2452	5	ATPases involved in chromosome partitioning-like	insertion in WH8109
ISL18	1859005	1863006	4001	2478	2480	3	PMT family 4-amino-4-deoxy-L-arabinose transferase/ glycosyltransferase, Photosystem II reaction centre protein PsbD/D2	different genes to CC9605, not defined as ISL in CC9605 ³
ISL19	1957096	1968729	11633	2589	2598	10	arginine biosynthesis bifunctional protein ArgJ, Dephospho-CoA kinase, Prolyl 4-hydroxylase, alpha subunit, glutamyl-tRNA(Gln) amidotransferase, B subunit, putative thiamine biosynthesis oxidoreductase, Nucleoside-diphosphate kinase, arginine decarboxylase	ISL in CC9605 contains different genes

¹WH8109 gene IDs are preceded by “Syncc8109_” in the GenBank file²Hypothetical genes and genes of unknown function are not listed³As per Dufresne et al. (2008)

Supplementary Table S14: Transcript levels of host tRNA genes. Log₂ fold change (FC) of infected versus control cultures with time (h) after infection. Positive and negative values indicate an increase and decrease in transcript levels during infection with significant differences in Log₂ FC (q<0.05) shown in red and blue, respectively.

Strain/Host	Gene name	product	anticodon	codon	Log ₂ FC (infected/control)						
					0h	0.5h	1h	1.5h	2h	3h	4h
Upregulated tRNAs											
WH8102	SYNW_RNA_8	tRNA-Ala	CGC	GCG	0.06	0.61	0.56	0.54	0.75	0.49	0.48
	SYNW_RNA_32	tRNA-Arg	TCT	AGA	0.45	1.05	1.26	1.26	0.65	1.24	1.42
WH7803	SynWH7803_RNA_8	tRNA-Ala	CGC	GCG	0.15	1.16	0.45	0.98	0.10	0.75	0.19
WH8109	Syncc8109_0300	tRNA-Ala	CGC	GCG	-0.30	-0.14	0.30	0.66	0.89	0.91	0.81
	Syncc8109_0158	tRNA-Arg	CCT	AGG	3.67	0.34	-1.10	-0.24	0.01	-0.20	0.14
Unchanged tRNAs											
WH8102	SYNW_RNA_23	tRNA-Ala	GGC	CCG	0.40	0.56	-0.22	-0.36	-0.30	0.25	0.38
	SYNW_RNA_4	tRNA-Arg	CCT	AGG	-0.01	0.01	0.52	0.07	-0.11	0.05	0.65
	SYNW_RNA_44	tRNA-Arg	CCG	CGG	-0.04	-0.15	-0.06	-0.34	-0.78	-0.48	0.23
	SYNW_RNA_6	tRNA-Arg	ACG	CGT	0.22	-0.16	-0.31	-0.41	-0.53	-0.21	0.14
	SYNW_RNA_54	tRNA-Asn	GTT	AAC	0.07	0.10	-0.28	-0.51	-0.42	0.02	0.37
	SYNW_RNA_30	tRNA-Asp	GTC	GAC	0.22	-0.15	-0.07	-0.17	-0.10	-0.28	0.03
	SYNW_RNA_43	tRNA-Cys	GCA	TGC	0.02	0.17	-0.25	-0.17	-0.02	-0.03	0.05
	SYNW_RNA_14	tRNA-Glu	TTC	GAA	0.15	0.12	0.42	0.29	-0.44	0.11	0.91
	SYNW_RNA_1	tRNA-Gly	CCC	GGG	0.52	0.39	-0.26	-0.26	-0.31	0.45	0.55
	SYNW_RNA_12	tRNA-Gly	TCC	GGA	0.35	-0.28	-0.09	0.11	-0.22	-0.72	0.05
	SYNW_RNA_2	tRNA-Gly	GCC	GGC	0.20	0.24	0.18	0.06	0.09	0.24	0.48
	SYNW_RNA_26	tRNA-His	GTG	CAC	0.09	-0.23	-0.11	-0.09	-0.37	0.02	0.42
	SYNW_RNA_22	tRNA-Leu	TAA	TTA	0.02	0.10	0.29	0.15	-0.15	0.01	0.46
	SYNW_RNA_25	tRNA-Leu	CAA	TTG	0.00	0.04	-0.01	0.26	-0.05	0.09	0.58
	SYNW_RNA_7	tRNA-Leu	TAG	CTA	0.01	-0.02	-0.48	0.10	-0.42	0.10	0.37
	SYNW_RNA_17	tRNA-Lys	TTT	AAA	0.32	0.00	-0.37	-0.51	-0.48	-0.42	0.35
	SYNW_RNA_20	tRNA-Met	CAT	ATG	0.19	-0.04	-0.71	-0.02	-0.08	0.13	0.50
	SYNW_RNA_28	tRNA-Ser	CGA	TCG	0.22	0.00	-0.06	-0.44	-0.40	-0.38	-0.30
	SYNW_RNA_13	tRNA-Thr	CGT	ACG	-0.05	-0.21	-0.75	-0.55	-0.30	-0.04	0.25
	SYNW_RNA_34	tRNA-Thr	GGT	ACC	0.22	0.05	-0.60	-0.52	-0.55	-0.03	0.17
	SYNW_RNA_41	tRNA-Thr	TGT	ACA	-0.24	0.02	-0.16	-0.13	0.47	0.44	0.75
	SYNW_RNA_11	tRNA-Val	GAC	GTC	-0.13	-0.05	0.26	0.02	0.03	0.28	1.29
	SYNW_RNA_31	tRNA-Val	CAC	GTG	-0.08	0.06	-0.46	-0.32	-0.07	0.38	0.69
	SYNW_RNA_55	tRNA-Val	TAC	GTA	0.39	0.46	0.04	-0.41	0.04	0.40	1.42
WH7803	SynWH7803_RNA_32	tRNA-Ala	GGC	GCC	0.06	0.24	-0.42	-0.14	-1.12	-0.25	-0.75
	SynWH7803_RNA_12	tRNA-Arg	CCG	CGG	0.13	0.20	-0.34	-0.16	-0.88	0.03	-0.78
	SynWH7803_RNA_4	tRNA-Arg	CCT	AGG	0.03	0.03	-0.10	-0.02	-0.13	0.02	-0.20
	SynWH7803_RNA_6	tRNA-Arg	CGT	ATG	0.11	-0.14	-0.55	-0.24	-0.90	-0.10	-0.46
	SynWH7803_RNA_52	tRNA-Asn	GTT	AAY	0.02	0.00	-0.15	-0.05	-0.18	0.14	-0.12
	SynWH7803_RNA_11	tRNA-Gln	TTG	CAA	0.03	0.08	-0.11	-0.08	-0.50	0.03	-0.39
	SynWH7803_RNA_2	tRNA-Gly	GCC	GGC	-0.03	0.29	-0.07	0.02	-0.42	-0.30	-0.24
	SynWH7803_RNA_1	tRNA-Gly	CCC	GGG	-0.02	0.10	-0.30	-0.09	-0.95	-0.23	-0.67
	SynWH7803_RNA_42	tRNA-Gly	TCC	GGA	0.04	0.04	-0.27	0.00	-0.68	-0.05	-0.34
	SynWH7803_RNA_35	tRNA-His	GTG	CAC	-0.03	0.04	-0.43	0.05	-0.50	0.12	-0.67
	SynWH7803_RNA_23	tRNA-Leu	TAA	TTR	0.22	0.67	0.23	0.40	-0.46	0.10	-0.47
	SynWH7803_RNA_7	tRNA-Leu	TAG	CTA	0.09	0.10	-0.46	-0.35	-0.87	-0.10	-0.71
	SynWH7803_RNA_33	tRNA-Leu	CAG	CTG	0.02	0.00	-0.22	-0.19	-0.37	-0.29	-0.29
	SynWH7803_RNA_46	tRNA-Leu	GAG	CTC	0.05	0.08	-0.29	-0.33	-0.74	-0.06	-0.50
	SynWH7803_RNA_27	tRNA-Lys	TTT	AAA	0.05	0.03	-0.34	-0.37	-0.68	-0.34	-0.48
	SynWH7803_RNA_29	tRNA-Met	CAT	ATG	0.01	0.05	-0.17	-0.02	-0.32	-0.06	-0.11
	SynWH7803_RNA_14	tRNA-Phe	GAA	TTC	0.03	0.21	-0.28	0.15	-1.03	-0.21	-0.56
	SynWH7803_RNA_24	tRNA-Pro	GGG	CCC	0.15	0.52	-0.14	0.20	-0.62	0.26	-0.32
	SynWH7803_RNA_28	tRNA-Pro	TGG	CCA	0.04	0.13	-0.08	0.14	-0.91	0.48	-0.45
	SynWH7803_RNA_9	tRNA-Pro	CGG	CCG	0.06	-0.11	-0.67	-0.17	-0.80	-0.20	-0.55
	SynWH7803_RNA_10	tRNA-Ser	GCT	AGC	-0.01	0.00	-0.15	-0.12	-0.66	-0.28	-0.41
	SynWH7803_RNA_15	tRNA-Thr	TGT	ACA	0.18	0.45	-0.25	0.30	-0.53	-0.28	-0.33
	SynWH7803_RNA_44	tRNA-Thr	CGT	ACG	0.06	0.36	-0.41	-0.13	-0.81	-0.19	-0.78
	SynWH7803_RNA_21	tRNA-Thr	GGT	ACC	0.04	0.44	-0.37	0.03	-0.83	-0.02	-0.57
	SynWH7803_RNA_40	tRNA-Val	CAC	GTG	0.00	0.16	-0.34	-0.14	-0.40	-0.17	-0.88
	SynWH7803_RNA_53	tRNA-Val	TAC	GTA	0.01	0.06	-0.34	-0.05	-0.43	-0.20	-0.48

					Log ₂ FC (infected/control)						
Strain/Host	Gene name	product	anticodon	codon	0h	0.5h	1h	1.5h	2h	3h	4h
WH8109	Syncc8109_1811	tRNA-Ala	GGC	GCC	0.18	-0.86	-0.29	0.07	0.08	0.32	0.35
	Syncc8109_2010	tRNA-Glu	UUC	GAA	1.40	-0.44	-0.65	0.11	0.30	-0.52	1.06
	Syncc8109_0049	tRNA-Gly	CCC	GGG	1.30	-0.93	-0.46	-0.01	-0.10	-0.18	1.01
	Syncc8109_1701	tRNA-Leu	UAA	TTA	2.09	-1.02	-0.48	0.23	0.31	0.16	0.43
	Syncc8109_1866	tRNA-Leu	CAA	TTG	0.66	-0.21	-0.60	-0.42	-0.06	-0.32	-0.01
	Syncc8109_0955	tRNA-Lys	UUU	AAA	0.64	-0.63	-0.83	-0.76	-0.50	-0.81	-0.03
	Syncc8109_0229	tRNA-Met	CAU	ATG	0.77	-1.21	-0.99	-0.94	-0.63	-1.13	-0.69
	Syncc8109_0334	tRNA-Pro	CGG	CCG	0.17	-0.61	-0.46	-0.21	-0.09	-0.24	0.09
	Syncc8109_1894	tRNA-Ser	CGA	TCG	1.02	-0.55	-0.95	-0.50	-0.50	-0.66	-0.49
	Syncc8109_0552	tRNA-Thr	GGU	ACC	0.50	-0.60	-0.33	-0.15	0.11	-0.37	-0.02
	Syncc8109_2235	tRNA-Val	GAC	GTC	2.30	-0.64	-1.10	-0.98	-0.20	-0.42	-0.21
	Syncc8109_2649	tRNA-Val	UAC	GTA	1.49	-0.54	-0.88	-0.48	-0.02	-0.40	0.28
tRNAs that cross hybridize with phage tRNA genes											
WH7803	SynWH7803_RNA_41	tRNA-Arg	TCT	AGA	0.07	0.16	0.20	0.62	-0.45	1.66	0.38
WH8109	Syncc8109_0624	tRNA-Arg	TCT	AGA	1.30	-0.05	-0.04	1.40	0.79	0.65	1.21